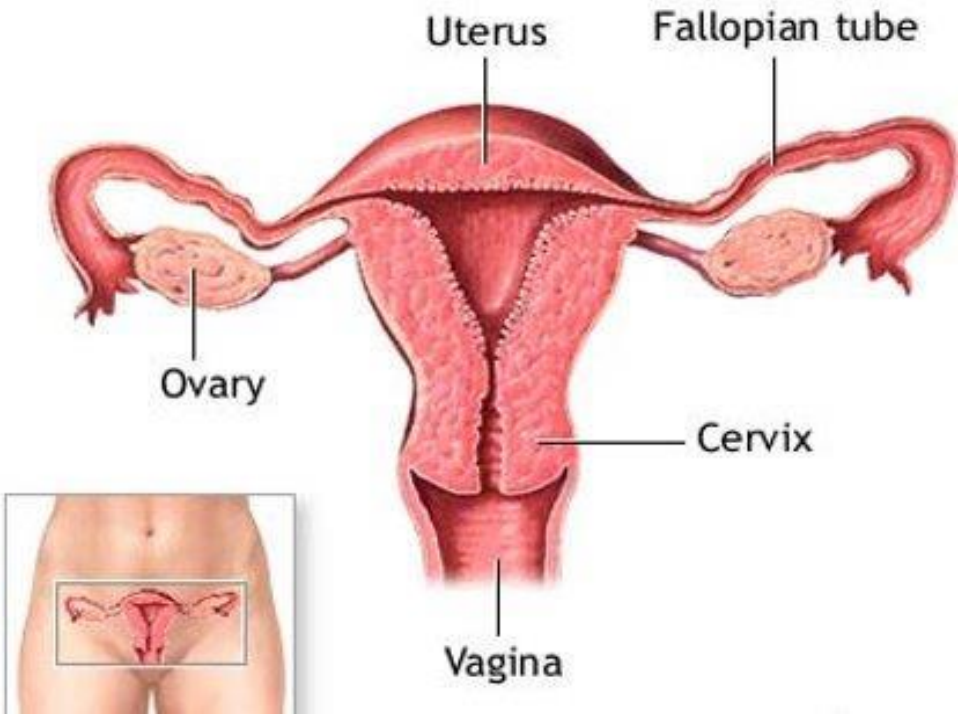
Physiology of the Ovary and Uterus



October 23, 2026

Katherine Lackritz, MD

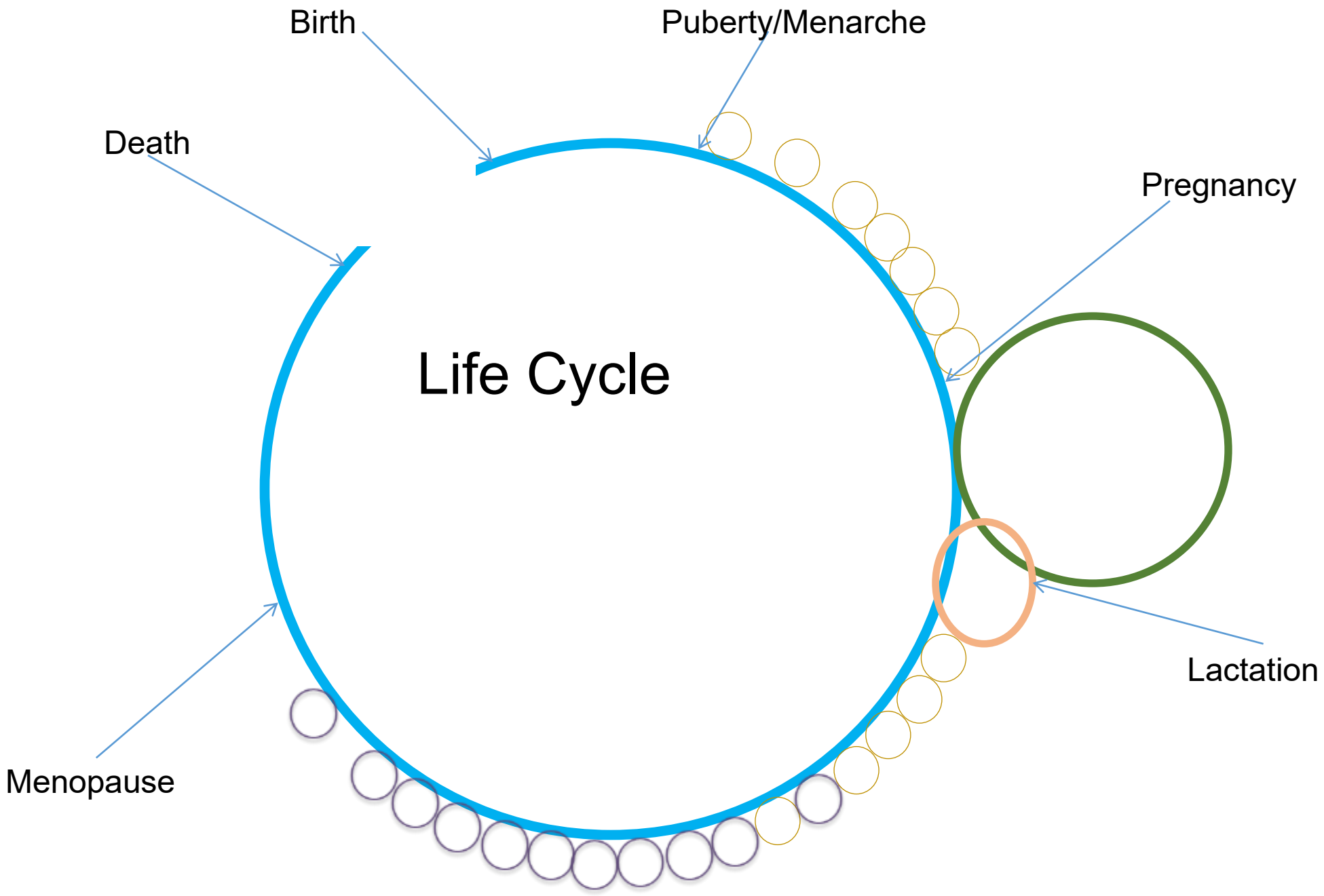
(Katherine.lackritz@jefferson.edu)

No disclosures

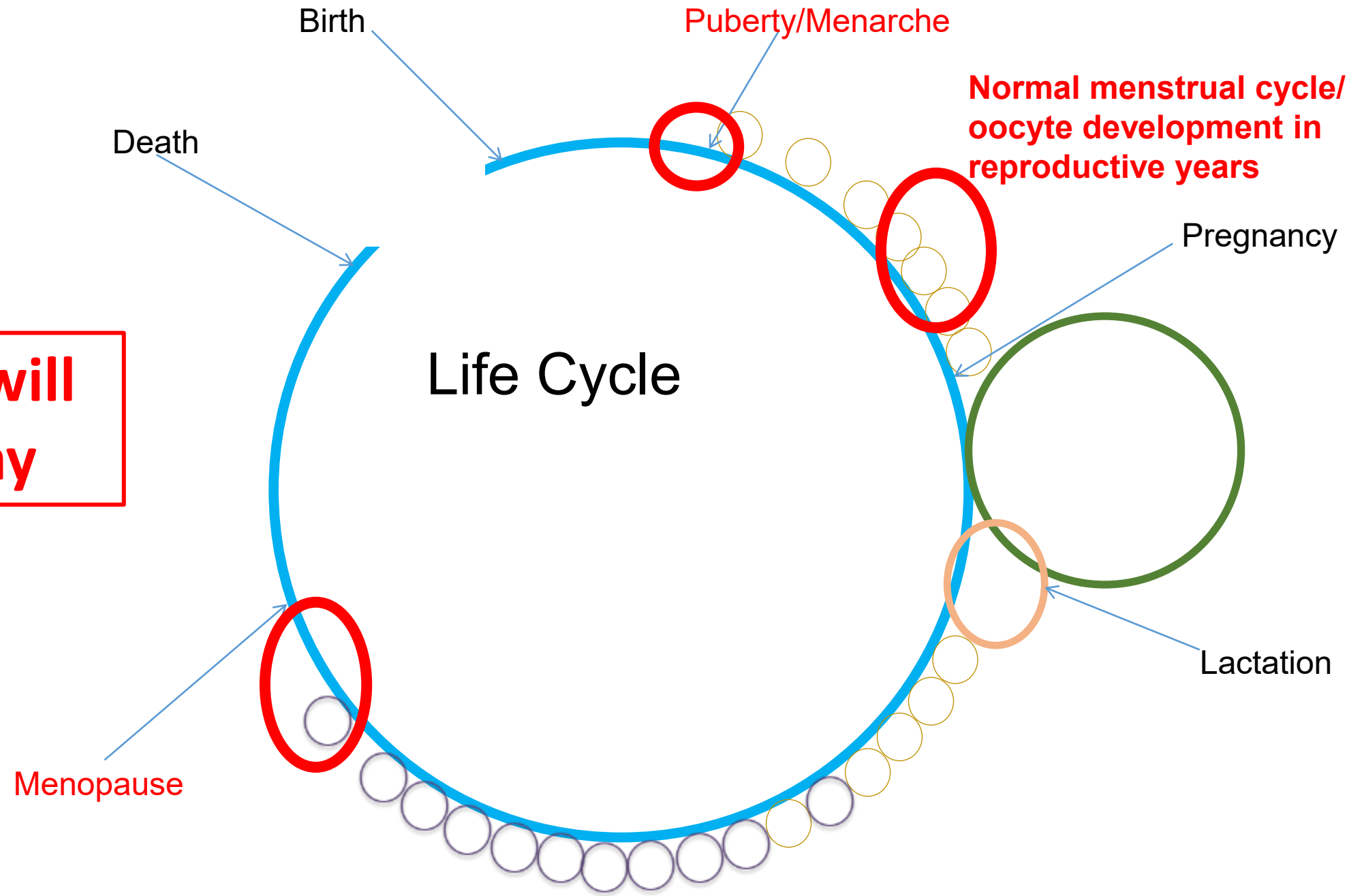
Learning Objectives

1. Recall the Hypothalamic-Pituitary-Ovarian axis and how puberty is initiated
2. Describe the phases of the menstrual cycle, including both ovarian (follicular) and uterine (endometrial) changes throughout the cycle
3. Describe the hormonal changes that happen throughout the menstrual cycle, focusing on LH, FSH, estradiol and progesterone
4. Explain the progression from primordial to dominant follicle and resultant ovulation and corpus luteum formation, including hormonal regulation
5. Summarize the process of meiosis in the oocyte and describe the process of fertilization
6. Describe the physiology of sexual response and sexual dysfunction
7. Define peri-menopause and menopause and describe the most commonly experienced symptoms

The Menstrual Cycle


Birth
Puberty/Menarche
Death
Pregnancy
Life Cycle
Menopause
Lactation

What we will cover today

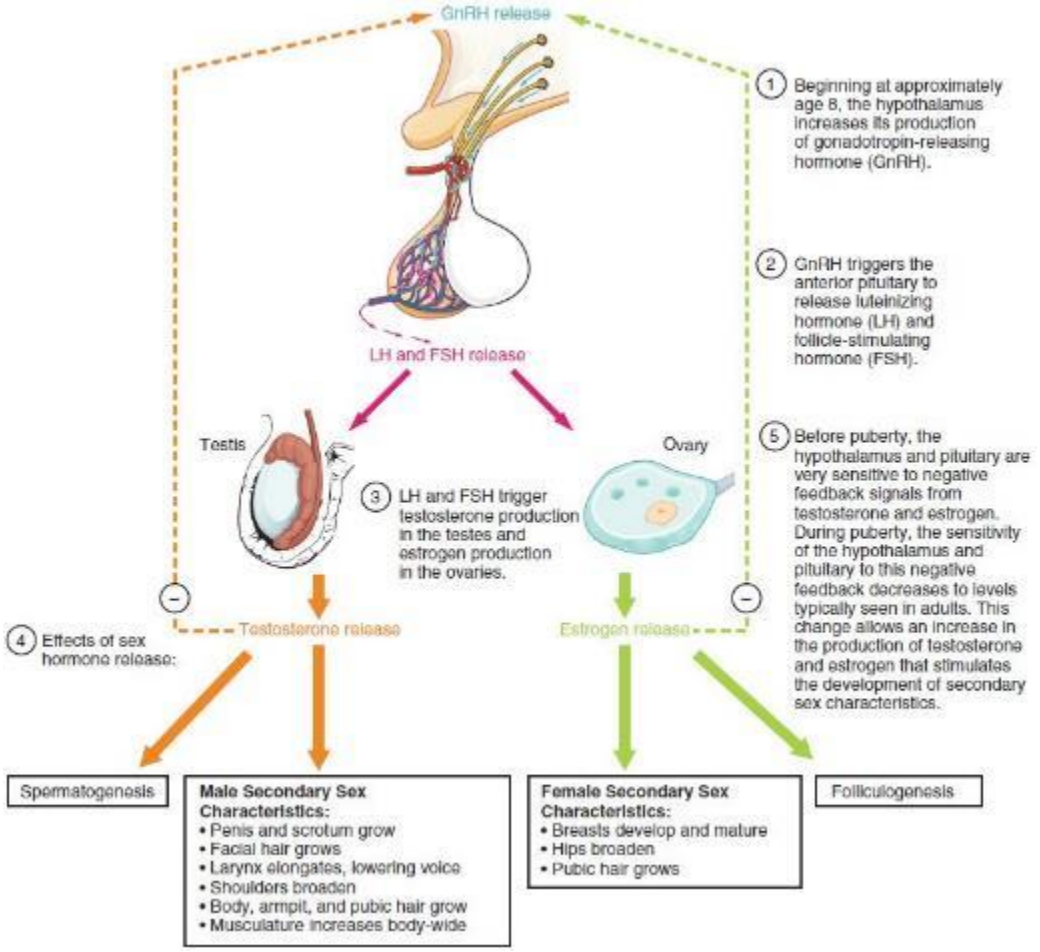


LO:

Recall the Hypothalamic-Pituitary-Ovarian Axis and how puberty is initiated

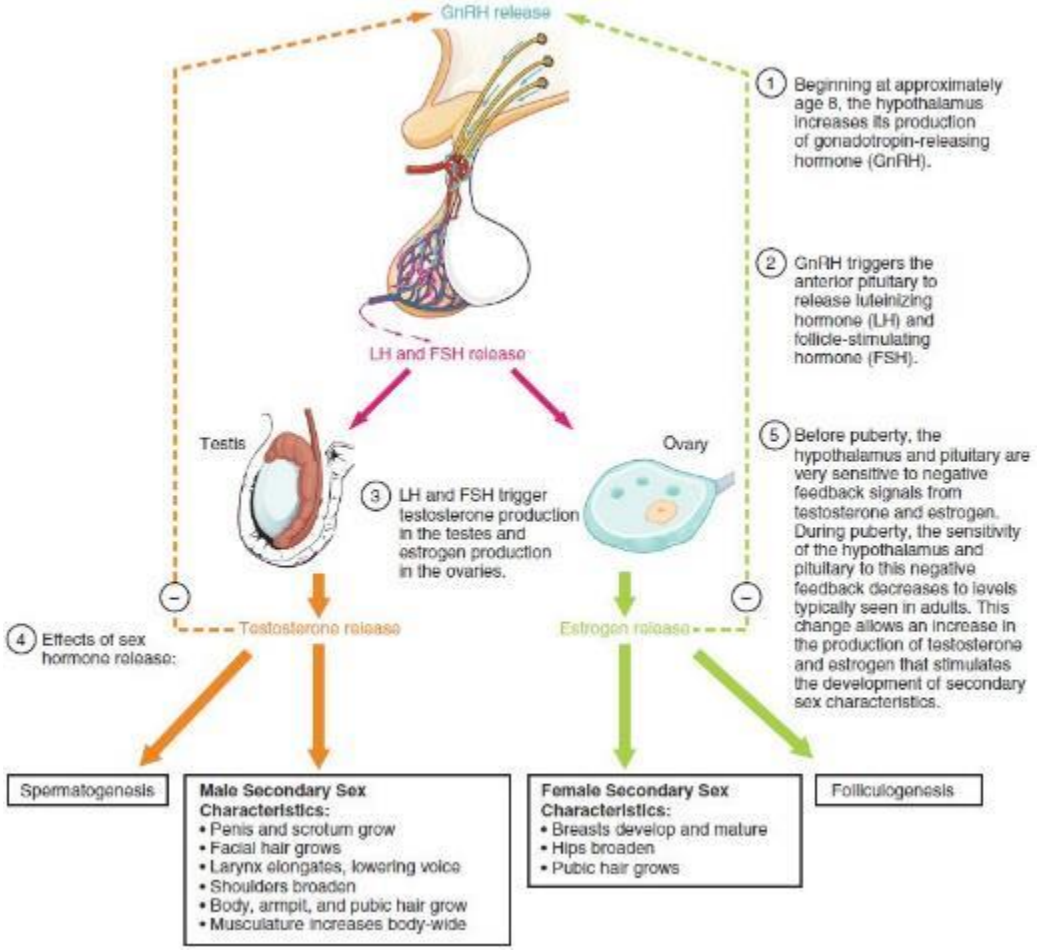
Puberty

- GnRH neurons develop and migrate to the hypothalamus in the fetus
- GnRH is suppressed in utero due to placental steroids
- After a baby is born its GnRH initially increases and then quiets down until puberty
- Puberty begins as HPO axis becomes reactivated, hypothalamus secretes GnRH pulses in increased amplitude and frequency
- Gonadotrophs in the anterior pituitary become **<u>more sensitive</u>** to GnRH



Puberty- GnRH pulses

- Increased GnRH leads to increased FSH and LH (gonadotropin) release from the anterior pituitary
  - FSH stimulates ovarian follicular growth
  - LH and FSH together stimulate production of estradiol by the ovaries
    - Estradiol stimulates breast development and growth spurt early in puberty
- Later in puberty, FSH and LH secretion by the pituitary and estradiol production by ovarian follicles leads to ovulation and menstrual cycles



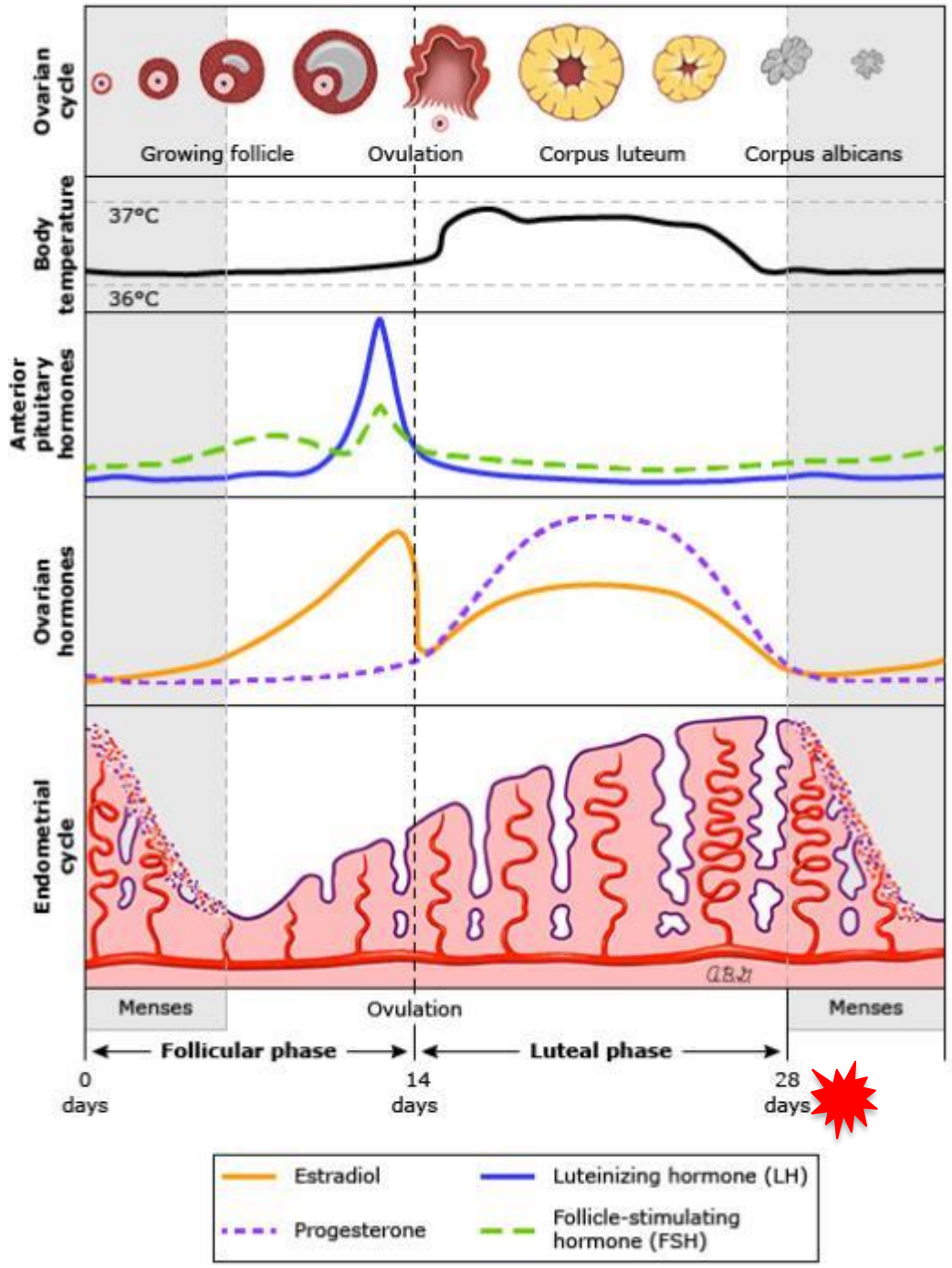
LO:

Describe the three phases of the menstrual cycle, including both ovarian (follicular) and uterine (endometrial) changes throughout the cycle

Menstrual Cycle- Three Phases:

Physiologic Goal is Reproduction

Menstrual Cycle Phase	Ovary	Endometrium
Follicular Phase (onset of menses to day before ovulation)	Estrogen Development/growth of dominant follicle	**Menses** (shedding) initially followed by estrogen driven thickening (**proliferative** phase)
Midcycle Surge/Ovulation	Ovulation (LH spike)	
Luteal Phase (ovulation through start of menses)	Progesterone production by corpus luteum	Maintenance and stabilization in preparation for embryo implantation (**secretory** endometrium)

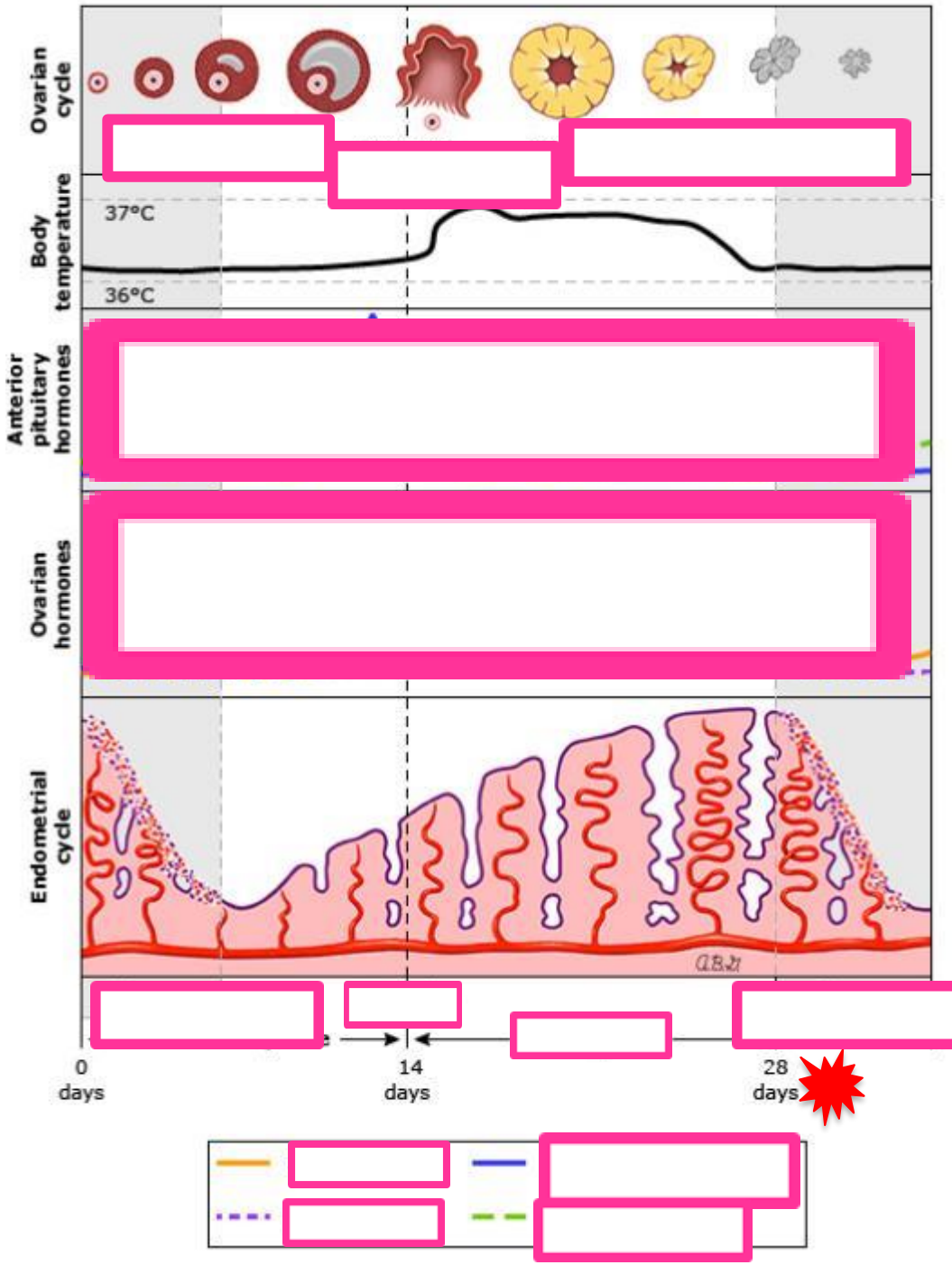


In this chart, day 28 is day 0 of the next cycle

Menstrual Cycle- Three Phases:

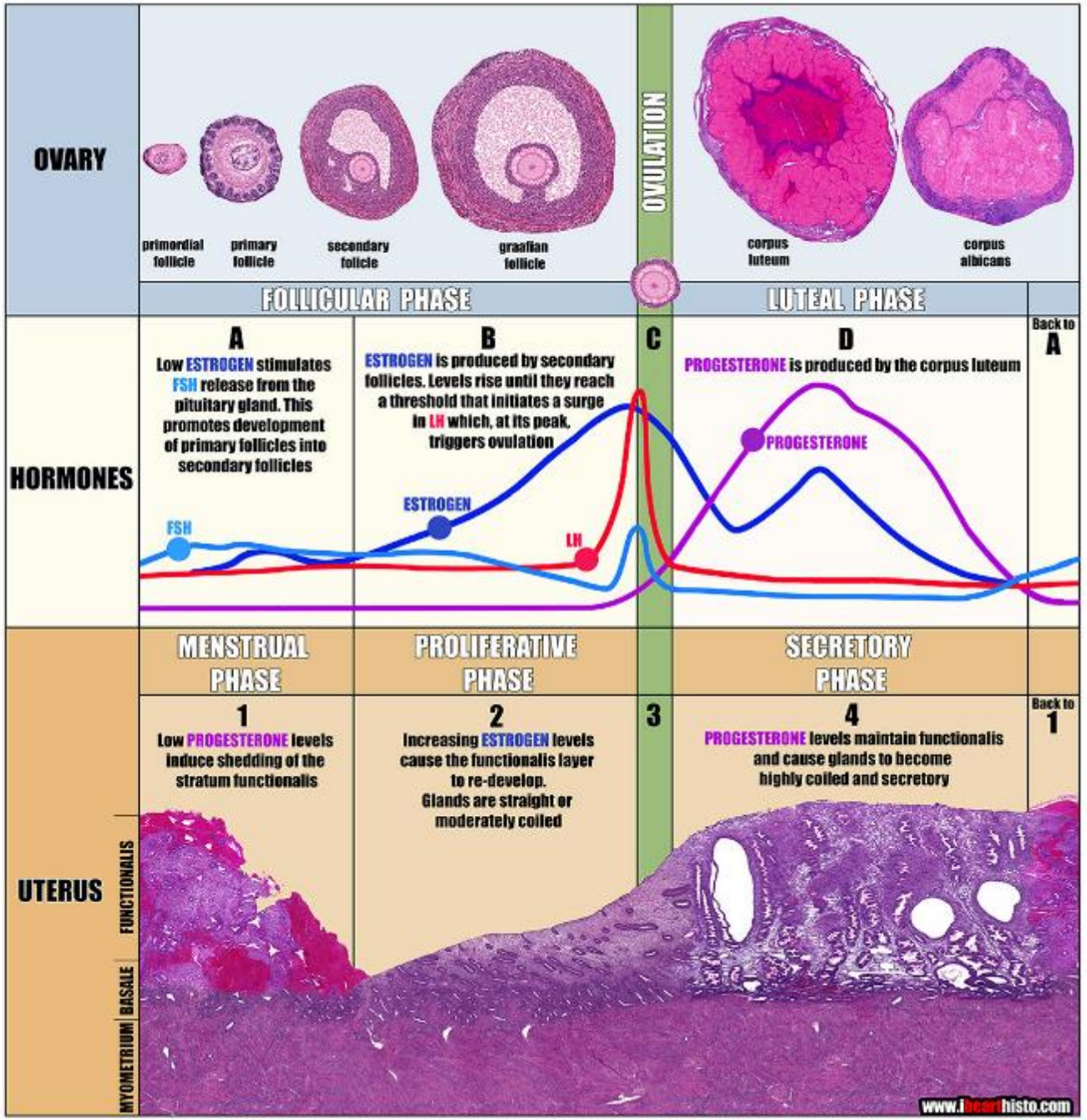
Physiologic Goal is Reproduction

Menstrual Cycle Phase	Ovary	Endometrium
Follicular Phase	Estrogen	**Menses proliferative**
Midcycle Surge/Ovulation	Ovulation (LH spike)	
Luteal Phase	Progesterone	**secretory**



In this chart, day 28 is day 0 of the next cycle

A little histology to bring it together!

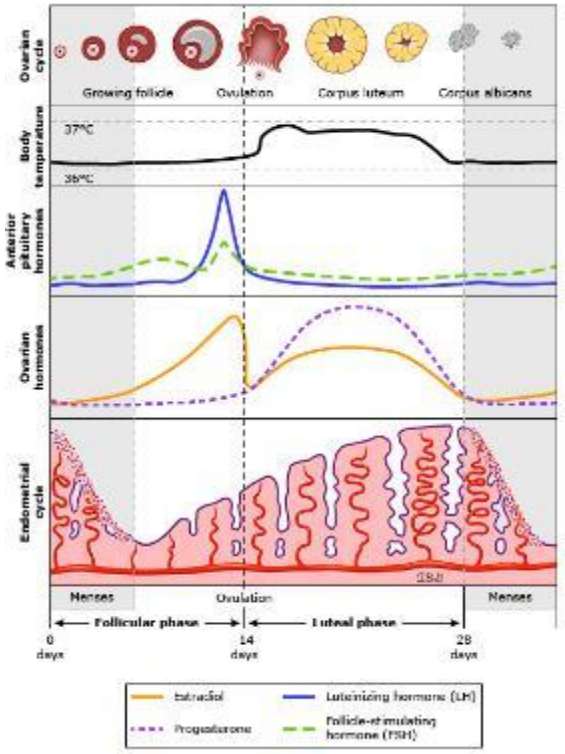


http://www.ihearthisto.com/post/119121528512/histology-of-the-menstrual-cycle-a-basic-review

Endometrial <u>histologic</u> phases:

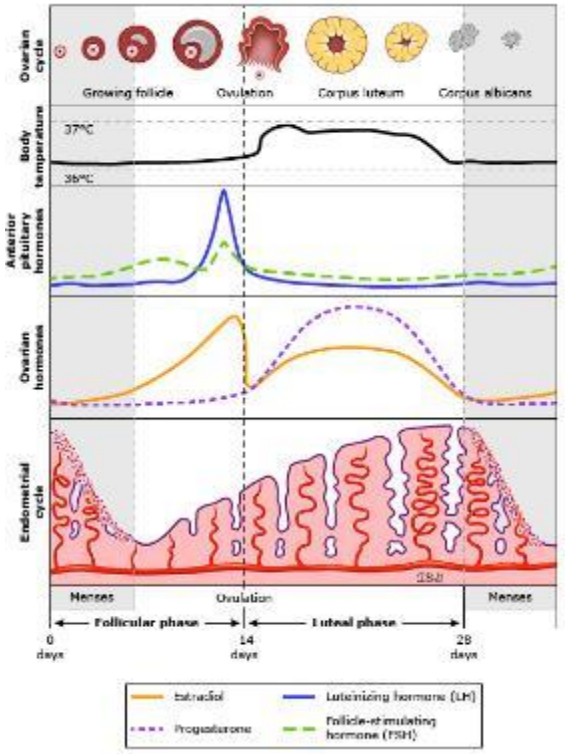
- The ovarian **follicular phase** includes both the uterine <u>menstrual</u> (shedding) and <u>proliferative</u> (thickening) phases
- The uterine <u>secretory phase</u> is during the ovarian **luteal phase**

Follicular Phase



- Lasts 10-16 days, dominant follicle develops in ovary and uterine endometrium sheds then proliferates
- Initially, during the luteal (previous cycle) to follicular transition:
  - Estradiol and inhibin drop, GnRH pulses are infrequent (1.5 to 2h).
  - FSH predominates
- As follicles grow, estradiol levels rise, FSH gets a little suppressed this leads to atresia of non-dominant follicles (fewer FSH receptors)
- During mid- to late-follicular phase, with increasing estradiol and inhibin A, a sudden switch to positive feedback of estradiol on the hypothalamus and pituitary occurs --> GnRH pulses occur more rapidly leading to midcycle LH spike

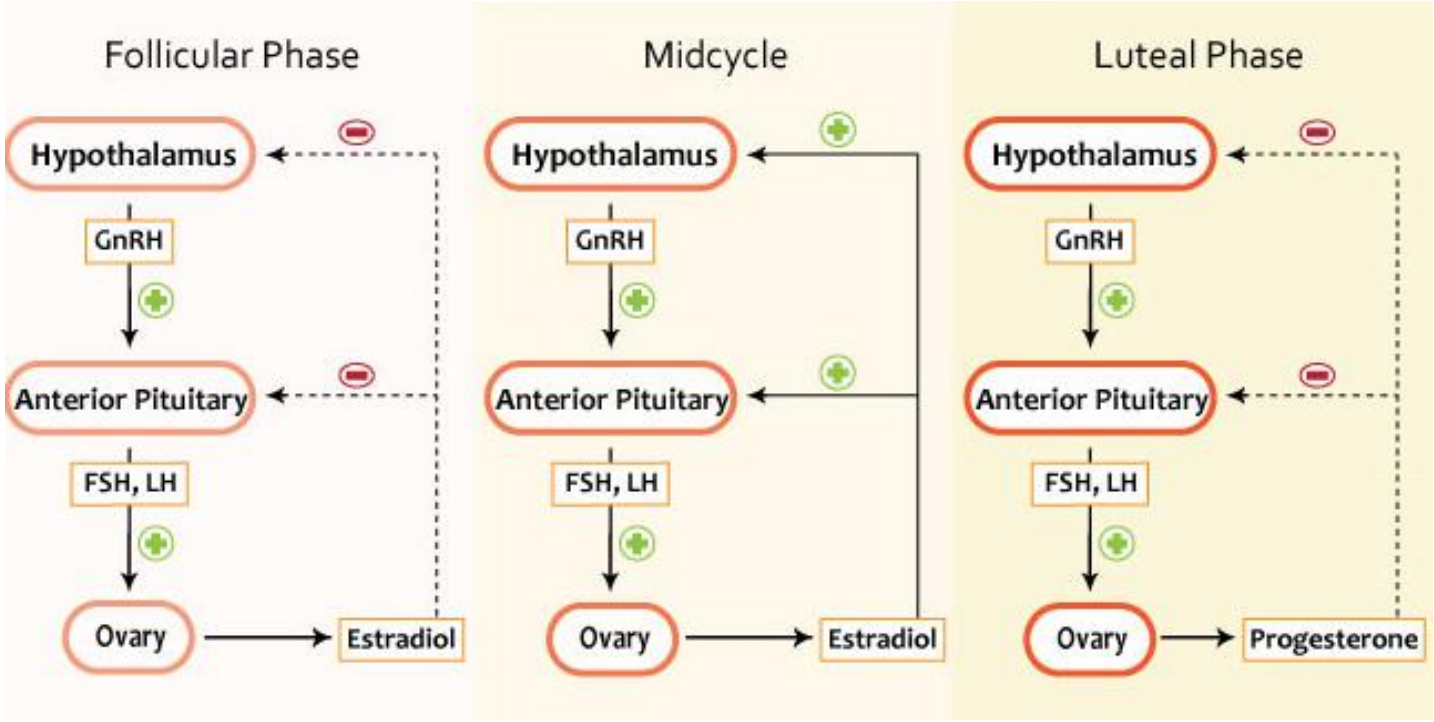
Follicular Phase

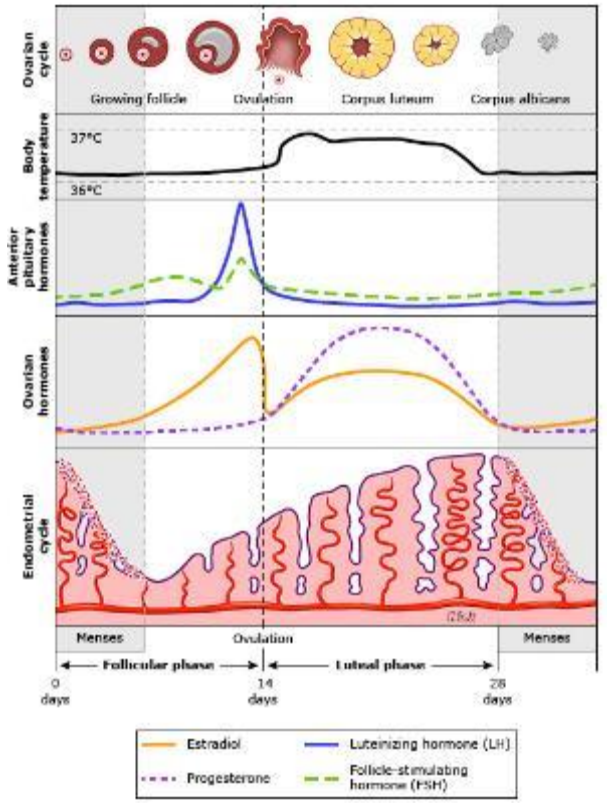


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From Dr. Haines' Hypothalamic-Pituitary Lecture

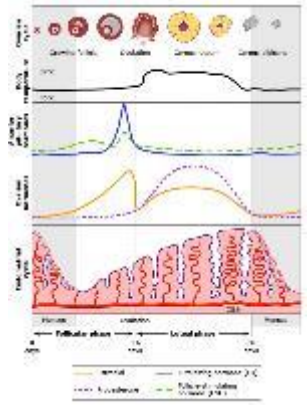
Hypothalamic-Pituitary-Ovarian Axis





LH surge --> ovulation

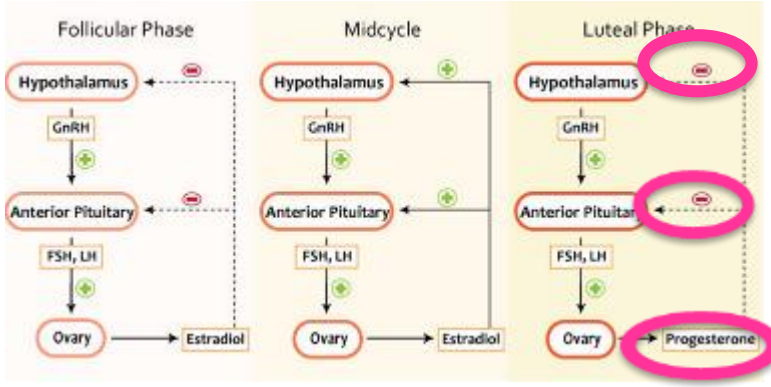
Ovulatory Phase



- Typically 14 days before menses, Lasts ~24 hours
- LH surge leads to resumption of meiosis in the oocyte and ovulation, leaving behind the corpus luteum
  - Egg is expelled from uterus into abdomen, hopefully picked up by fallopian tube (clinical correlate: ectopic pregnancy)



- Estradiol must be maintained at a critical concentration for a specific amount of time prior to the LH surge- any reduction will stop ovulation (follicle failure --> anovulation)
- In the presence of progesterone, the surge will also not occur
  - Clinical correlate: this is how "Plan B" emergency contraception works!

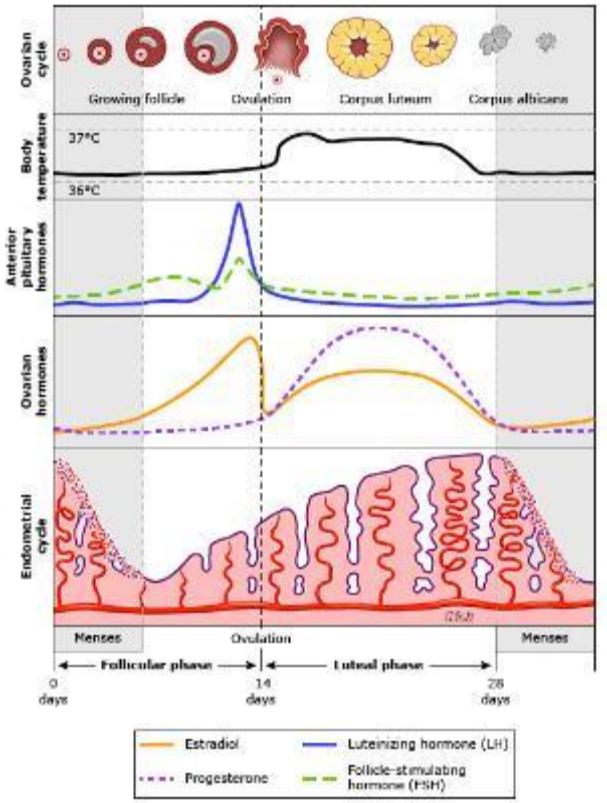


Luteal phase

- https://www.tiktok.com/@actressbecc/video/7489215707104939310
- https://www.tiktok.com/@greysonhoelzel/video/7407577922812333354

Luteal phase

- Progesterone dominates
- Endometrium becomes secretory (under the influence of progesterone), preparing for the implantation of an embryo
- LH maintains corpus luteum
- As progesterone levels inevitably fall with failure to conceive the endometrium loses its support and starts to become ischemic



The Ovary

- Major Functions
  - Steroidogenesis
    - Estrogen
    - Progesterone
    - Androgens
  - Maturation of germ cells
- Each cycle one follicle becomes dominant
- Dominant follicle releases egg
- Other follicles undergo atresia
- Ruptured follicle becomes corpus luteum which secretes progesterone

LO:

Describe the hormonal changes that happen throughout the menstrual cycle, focusing on LH, FSH, estradiol and progesterone

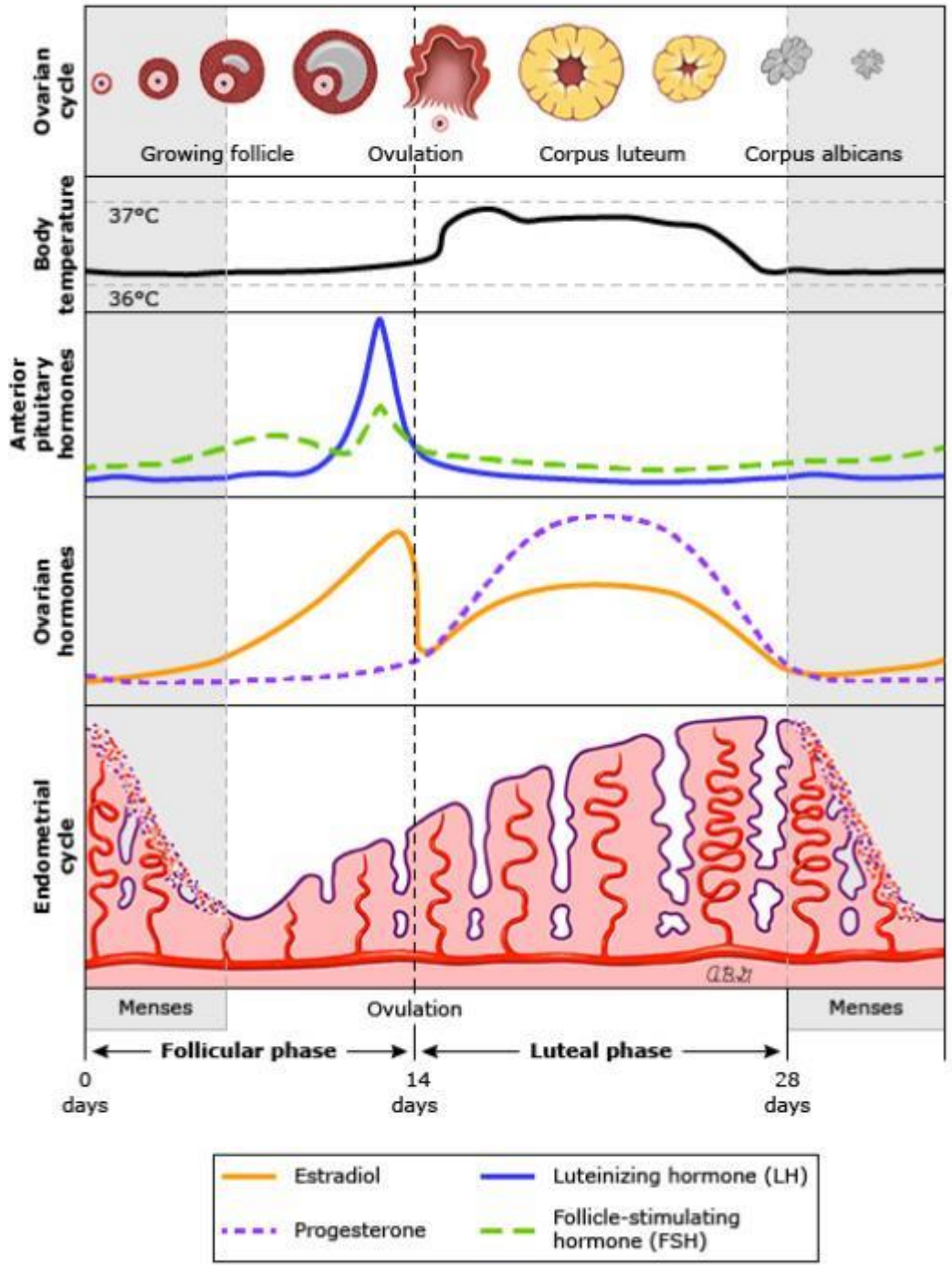
Menstrual Hormones

Ovarian

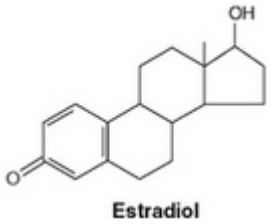
- **Estradiol**
- **Progesterone**

Anterior Pituitary

- **Follicle Stimulating Hormone (FSH)**
- **Luteinizing Hormone (LH)**



Estradiol



Follicular phase

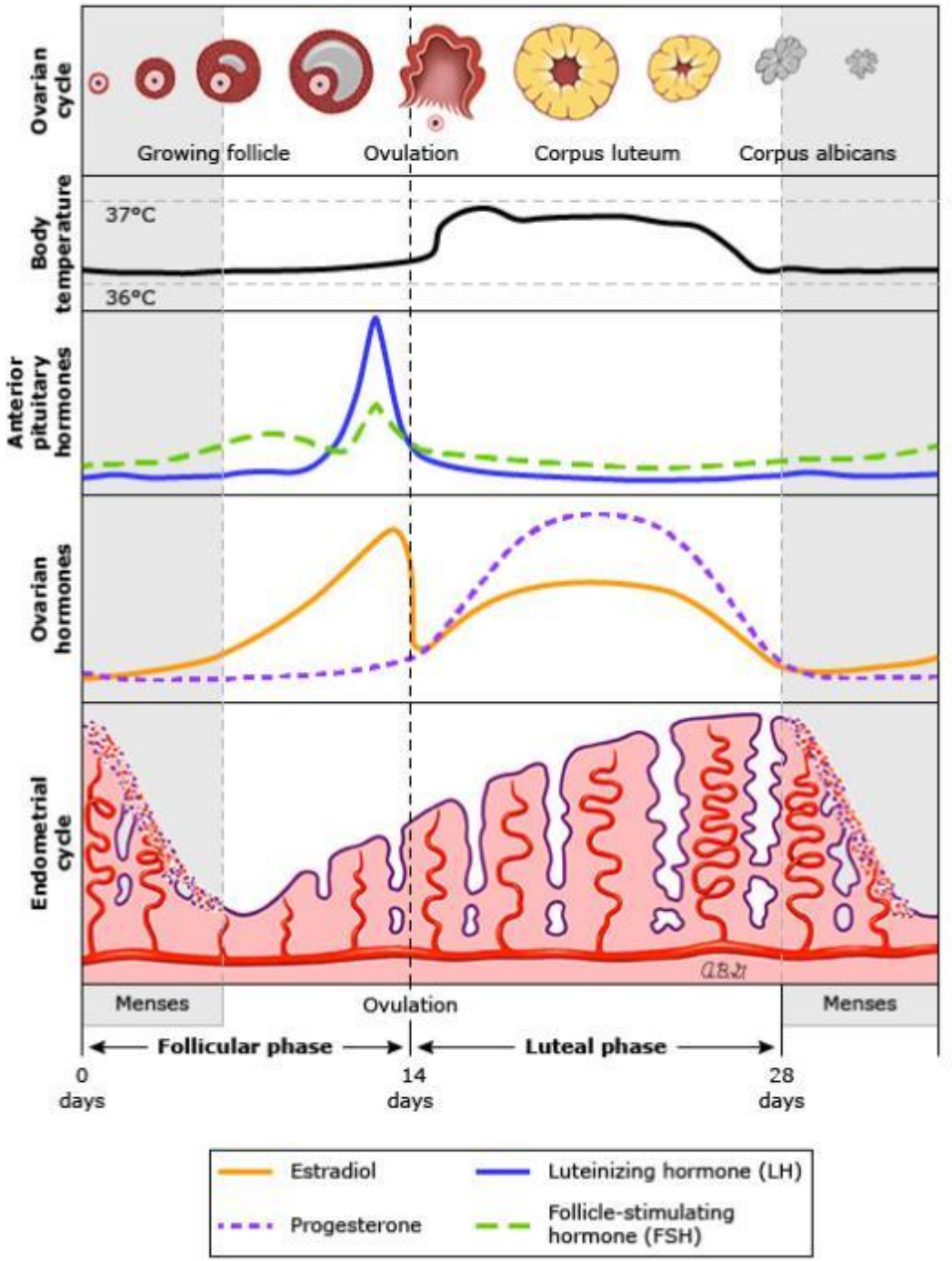
- Estradiol levels are low at the start of the cycle
- The growing cohort of follicles make increasing amounts of estradiol due to growth/division of theca and granulosa cells.
- The endometrium is thickening and increasing in glandularity in response to estradiol

Midcycle/ovulation

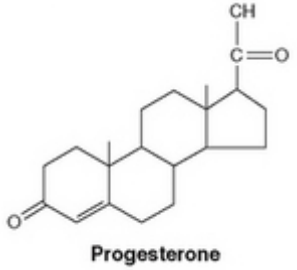
- The LH surge triggers ovulation (~12-24h after LH peak) and the mature oocyte extrudes from the follicle, leaving behind the corpus luteum.
- Estradiol levels drop around ovulation – clinical pearl, can cause "mid cycle spotting" for some people with a uterus

Luteal phase

- Estradiol levels increase again, produced by corpus luteum .
  - If pregnancy does not occur, corpus luteum dies off and estradiol levels drop
  - If pregnancy occurs, human chorionic gonadotropin (HCG) supports the corpus luteum



Progesterone



Follicular phase

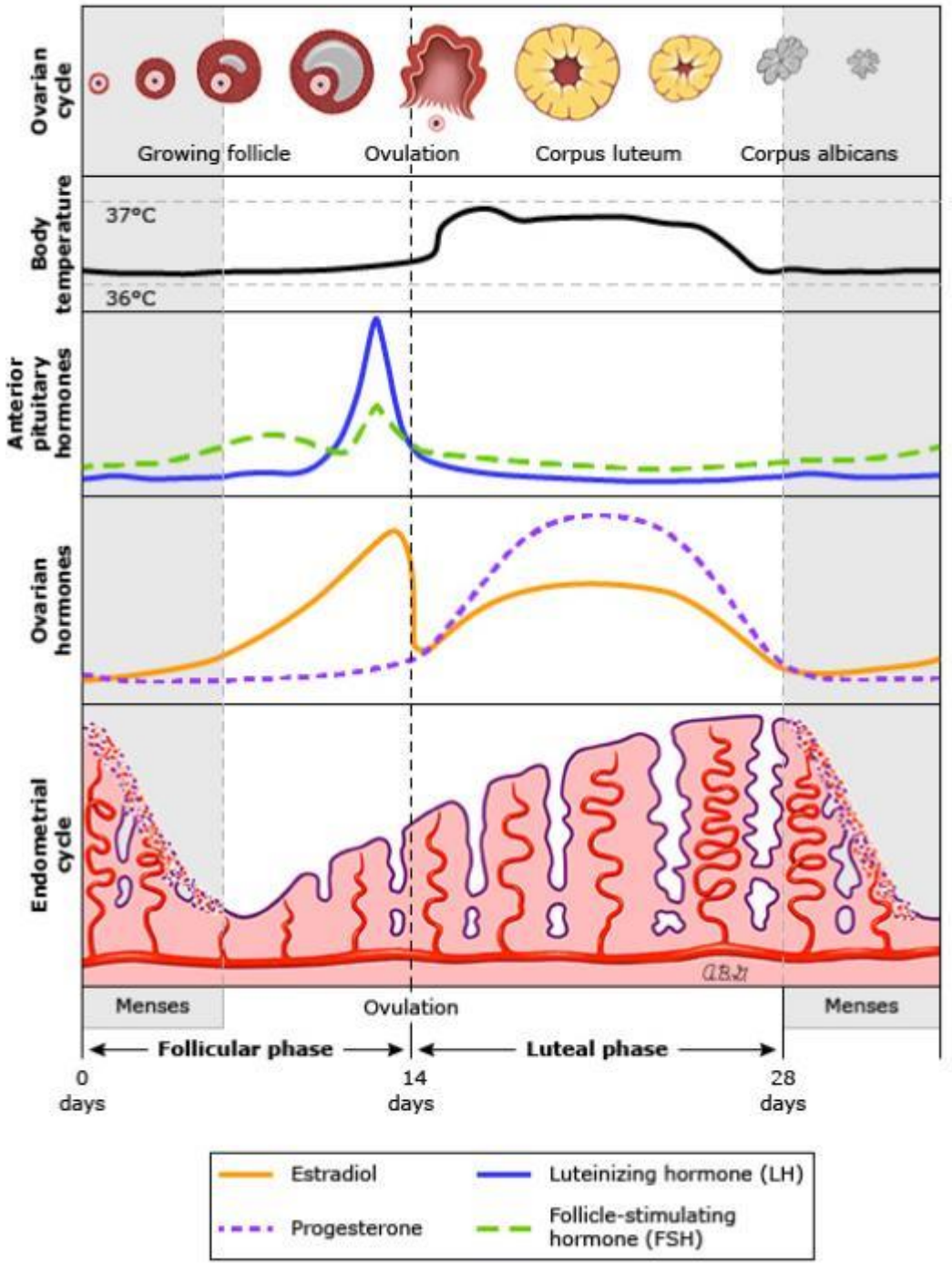
- Progesterone levels are also low at the start of the cycle and remain low until after ovulation

Ovulation

- Mature oocyte ovulates and corpus luteum is left behind
- Granulosa cells luteinize and produce progesterone

Luteal phase

- The corpus luteum makes high amounts of progesterone
- In the endometrium, cessation of mitoses and stabilization, gland organization, spiral artery development, and maintenance is regulated by progesterone
- If pregnancy does not occur, the corpus luteum dies off and progesterone levels drop, leading to destabilization of the endometrium and menses



Progesterone clinical pearls

- The presence of systemic progesterone negatively inhibits pituitary hormones that cause ovulation- this is how systemic hormones stop ovulation for contraception
  - "the pill" is estrogen and progesterone together
  - Nexplanon- systemic progesterone
  - Depo-provera- IM 12 week progesterone
- Local progesterone- IUD in uterus (Mirena, Kyleena, Lileta…)
  - Not seen systemically, 85% of people still ovulate
  - Local effect makes cervical mucus very difficult for sperm to navigate
- Anovulatory people with uteruses have "unopposed estrogen" and can have endometrial over-proliferation, can lead to issues

Follicle Stimulating Hormone (FSH) and Luteinizing Hormone (LH)

Follicular phase

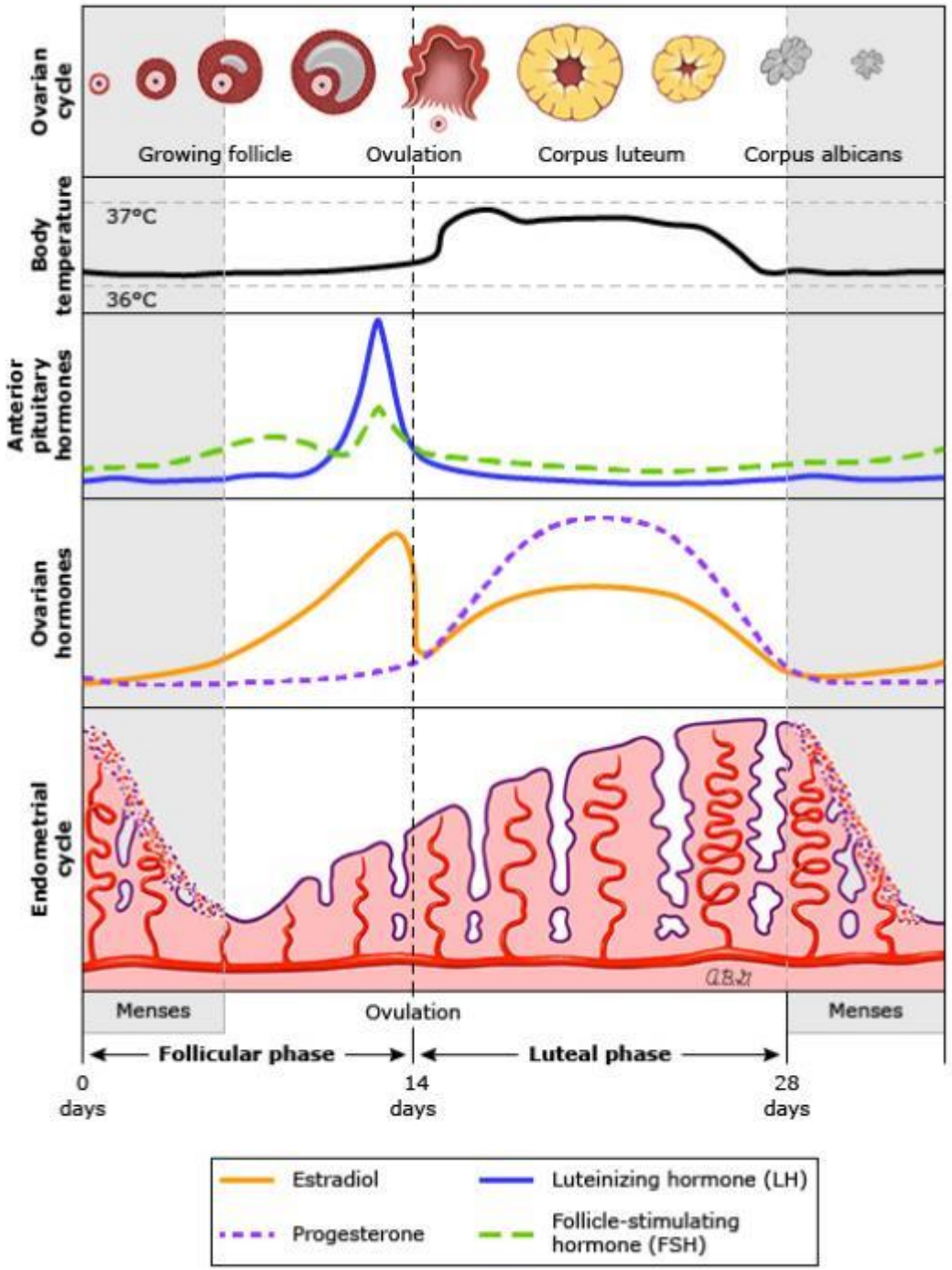
- Drop in E/P from last cycle -> decrease in negative feedback on hypothalamus -> increased GnRH pulse frequency -> FSH increase.
- FSH recruits a cohort of follicles, one of which will become dominant. Follicles make estradiol (and inhibin A), which suppress FSH and LH
- As the dominant follicle is selected, FSH induces ovarian LH receptors

Midcycle/ovulation

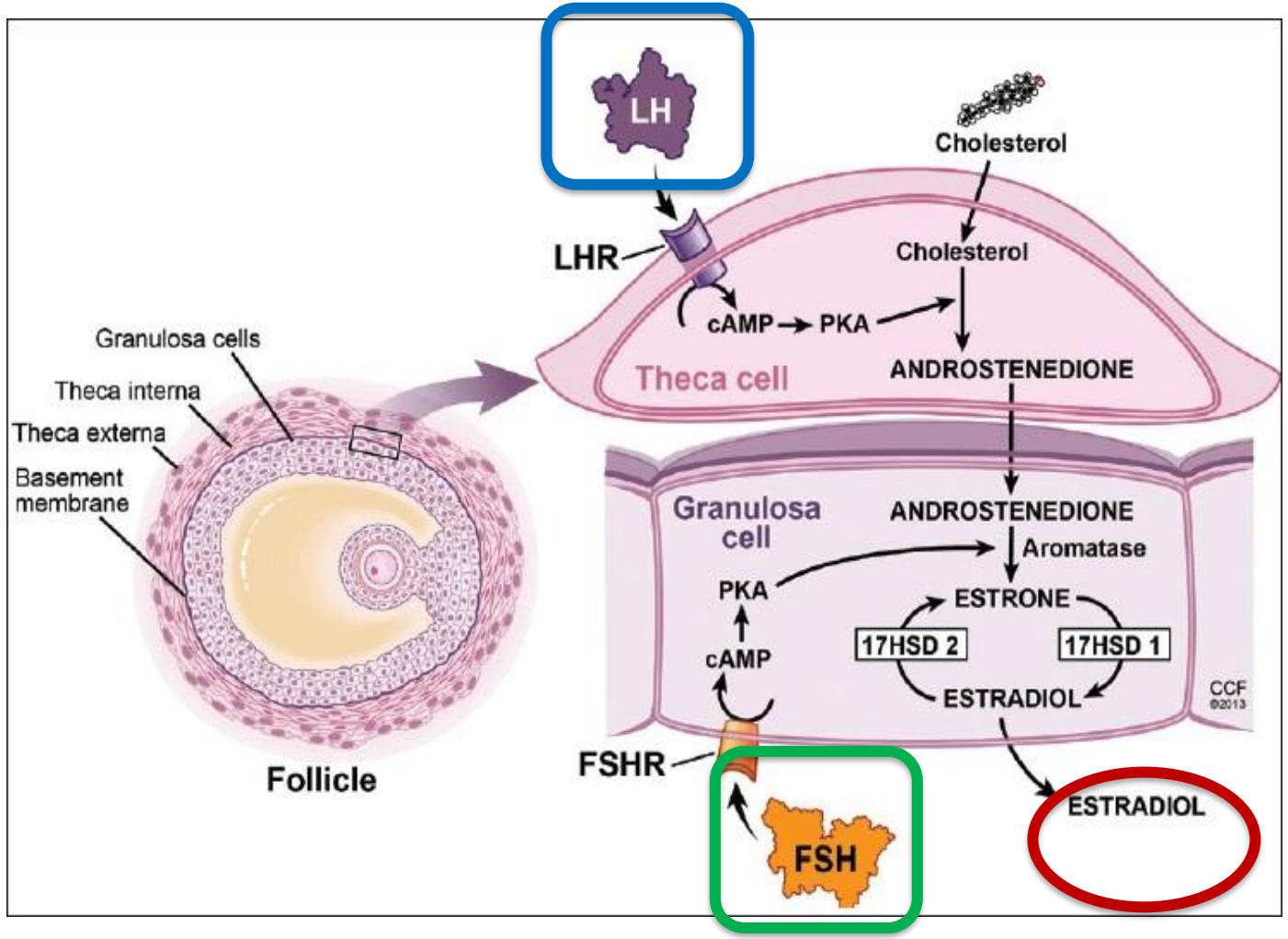
- **Suddenly**, midcycle, estrogen switches to a positive feedback effect leading to spikes of both FSH and **LH (pulsatile)**. This feedback switch is poorly understood.
- LH surge causes the dominant oocyte to complete its first meiotic division (required for fertilization). The oocyte is released (ovulation)!!

Luteal phase

- Switch back to negative feedback of estradiol and progesterone on hypothalamus (GnRH) leading to low FSH and LH levels
- No implantation -> corpus luteum dies -> estradiol, progesterone and Inhibin A levels fall -> gradual increase in GnRH and rise in FSH, initiating recruitment of the next cohort of follicles!



Estradiol and Progesterone Production in the Ovarian Follicle: Both LH and FSH Dependent



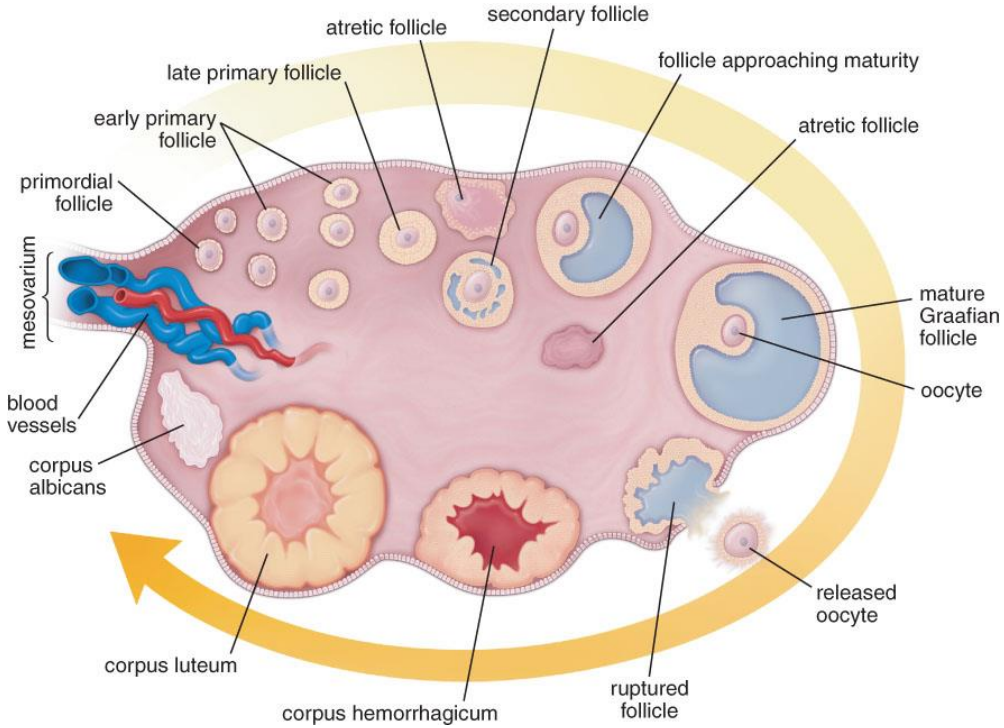
- LH acts on theca cells to produce androgens
- Androgens diffuse to granulosa cells
- In granulosa cells, androgens are converted to estrogens via FSH supported aromatization
- As a follicle matures it has an increasing number of LH and FSH receptors -
  > increased estradiol production
- Granulosa cells luteinize around ovulation to become granulosa lutein cells (in corpus luteum) and begin to produce progesterone in addition to estradiol

LO:

Explain the progression from primordial follicle to dominant follicle and resultant ovulation and corpus luteum formation, including hormonal regulation

Follicular Development in the Ovary

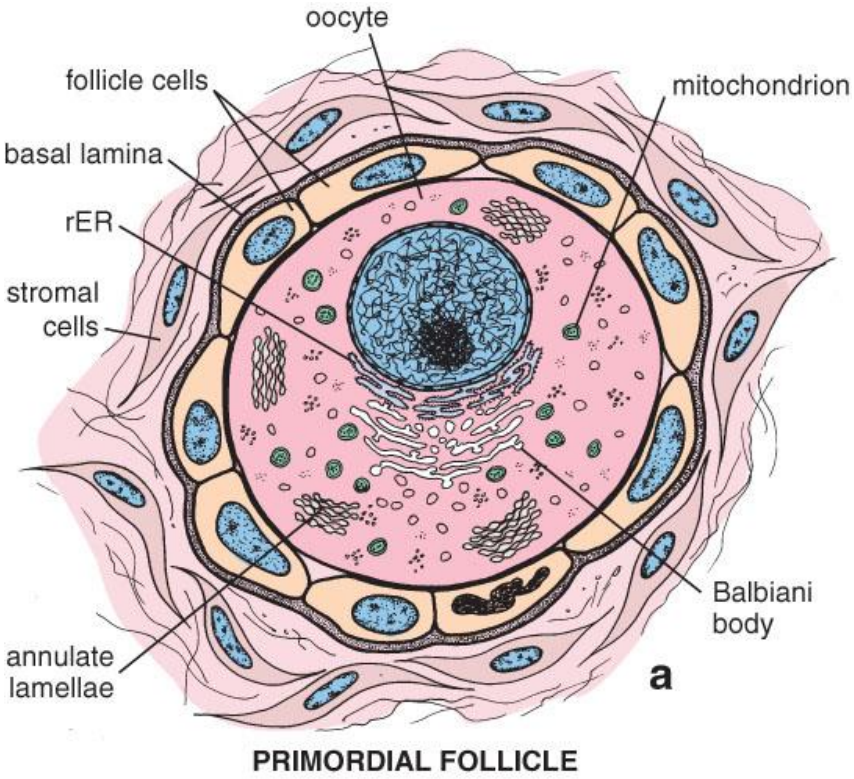
- Primordial follicle
- Growing follicles
  - Primary follicle
  - Secondary follicle (aka antral follicle)
- Graafian follicle (mature follicle)



Follicular Development

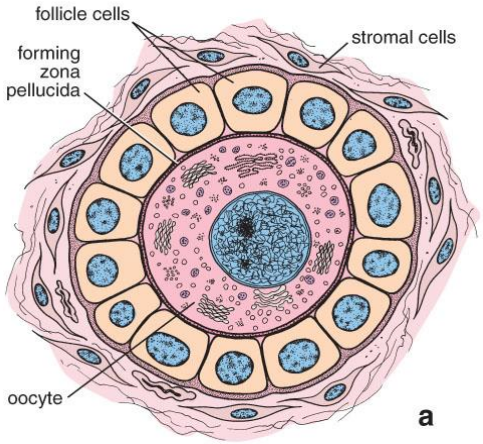
- **<u>Primordial follicle:</u>** formed in the human fetus (in utero); we are born with all our oocytes that are capable of reproduction
  - Made up of a single primary oocyte surrounded by a single layer of flat granulosa cells and a basement membrane
  - Primary oocytes began meiosis 1 but arrested at Prophase 1
    - We'll get to this again in the next LO
  - No independent blood supply, minimal endocrine influence
  - Over the years many are lost to apoptosis, and some are "recruited" to be primary follicles, until very few/none are present after menopause
  - The number left constitutes the "ovarian reserve"

- Primary follicle
- Secondary follicle (aka antral follicle)
- Graafian follicle

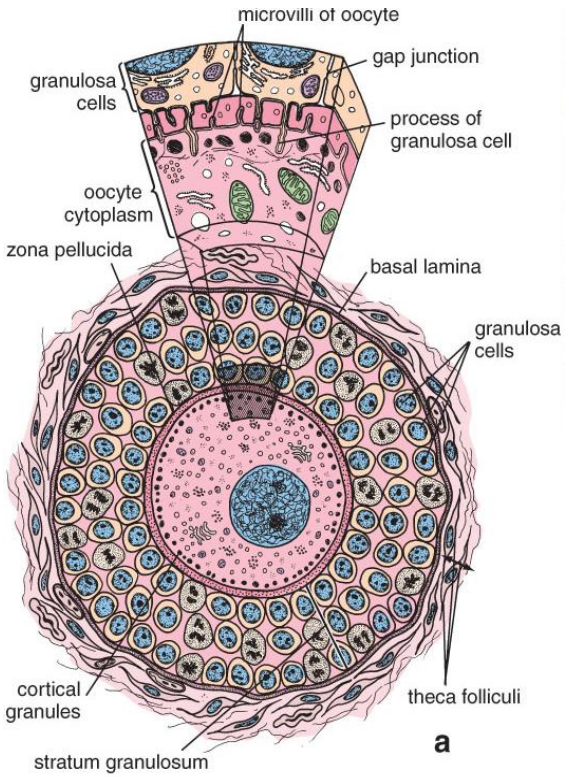


Follicular Development

- Primordial follicle
- **Primary follicle:** recruited from the primordial follicle reserve relatively constantly for the first 30 years of life
  - Various growth factors control recruitment, including many secreted by the oocytes themselves, not yet dependent on gonadotropins
  - Granulosa cells become cuboidal followed by growth and differentiation of the oocyte
  - FSH receptors begin to be expressed on granulosa cells; necessary for further follicle development
  - Zona pellucida (extracellular matrix) forms around oocyte- protects the oocyte and plays a role in fertilization by sperm
  - Intercellular connections between oocyte and granulosa cells develop allowing regulatory molecule and nutrient passage
  - Change in granulosa cells from simple cuboidal to stratified and proliferation
  - Increased gap junctions between granulosa cells for improved communication
  - Theca cell layer forms outside the basement membrane
  - Angiogenesis occurs bringing circulation, allowing nutrients, gonadotrophins and waste to pass
  - Overall, contains a fully grown oocyte with zona pellucida, multiple layers of granulosa cells, basement membrane, and theca layer (or two) with capillary net
  - Meiosis continues to be inhibited – still stuck at Prophase 1
- Secondary follicle (aka antral follicle)
- Graafian follicle



PRIMARY FOLLICLE



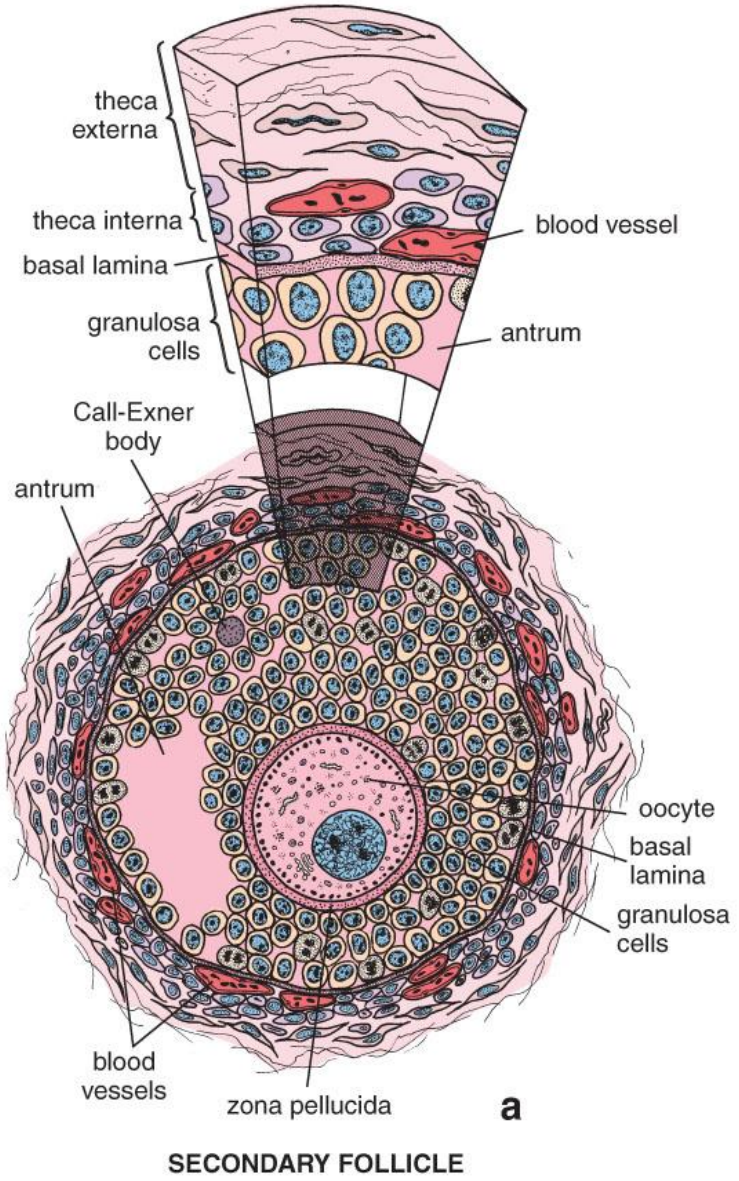
LATE PRIMARY FOLLICLE

Follicular Development

- Primordial follicle
- Primary follicle

- **<u>Secondary follicle (antral follicle):</u>**
  - First time we see follicular fluid (in the antrum- at one pole)
    - Plasma exudate from oocyte and granulosa cells
    - Regulatory molecules pass through this fluid
    - Occurs when stratum granulosum reaches 6-12 cell layers
  - Measure 0.4 to 25mm in diameter and can be seen by ultrasound
    - Clinical correlate: "antral follicle count" used as part of fertility eval (proxy for ovarian reserve) and treatment planning
  - Enormous growth, but structure remains largely the same
  - Theca divides into two layers (externa and interna)

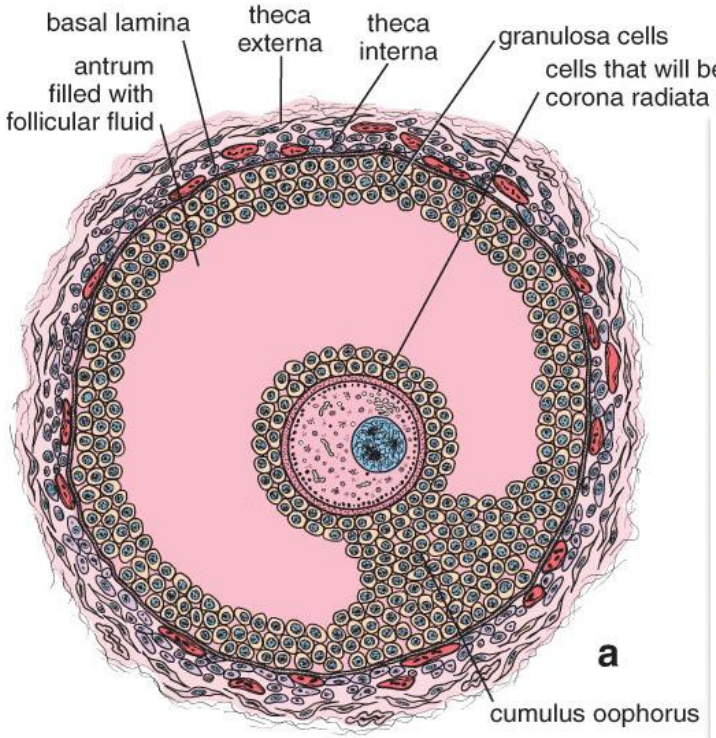
- Graafian follicle



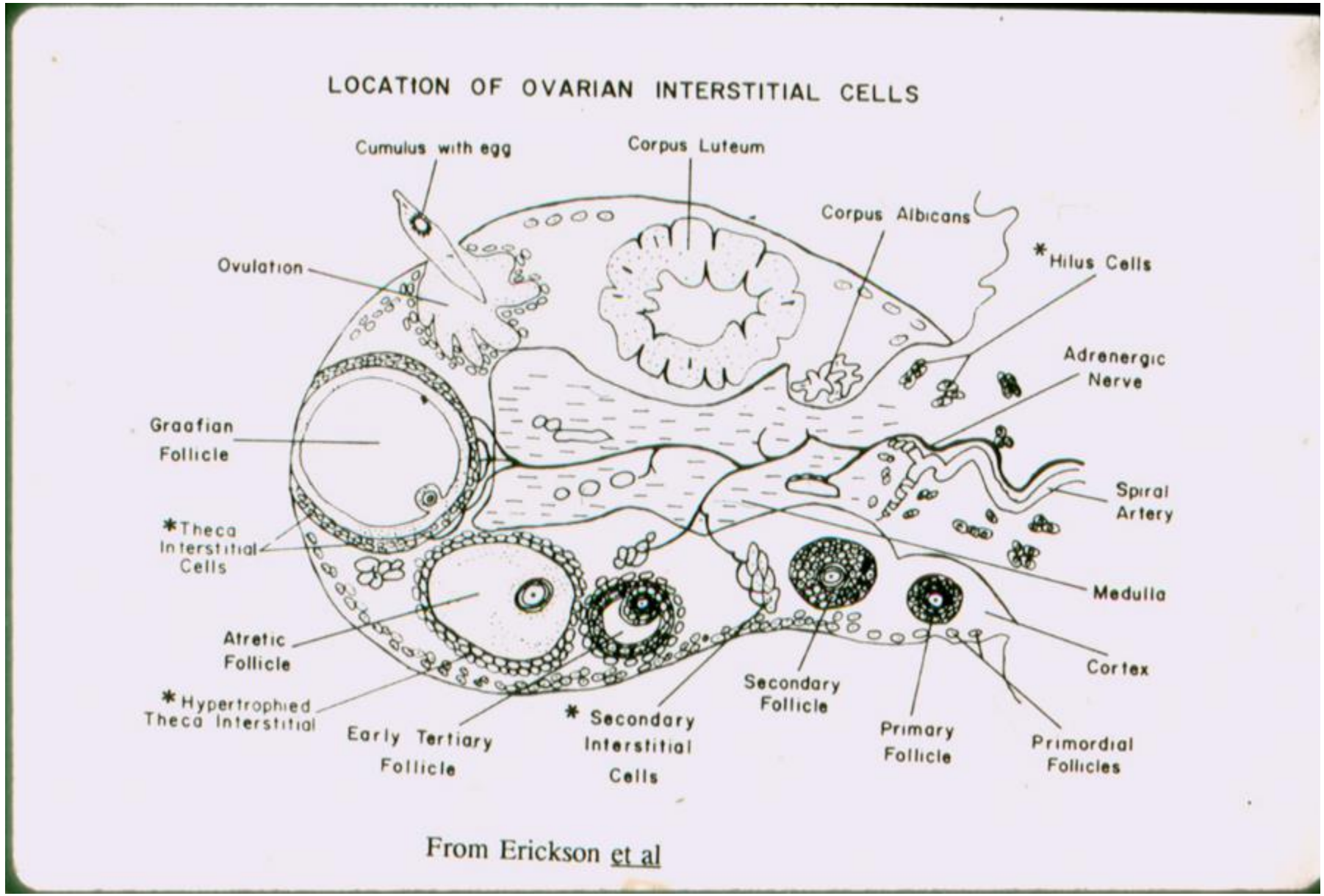
Follicular Development

- Primordial follicle
- Primary follicle
- Secondary follicle (aka antral follicle)

- **<u>Graafian follicle:</u>**
  - Approximately 5-10 Graafian follicles begin to mature each menstrual cycle; one eventually becomes the dominant follicle, and the rest become atretic
  - Granulosa cells (FSH responsive) and theca cells (LH responsive) continue to proliferate, and antrum continues to fill with fluid
  - The follicle that becomes dominant has as much as 100-fold proliferation of granulosa and theca cells
    - At ovulation, measures 2.5 + cm and contains about 50 million granulosa cells and 7mL of follicular fluid!
  - Theca externa differentiates into smooth muscle cells
  - Theca interna become theca interstitial cells (steroid-producing) that have insulin and LH receptors and a rich capillary network.
    - Produce high levels of androgens with LH/insulin stim
    - Clinical correlation: PCOS! (androstenedione --> testosterone)

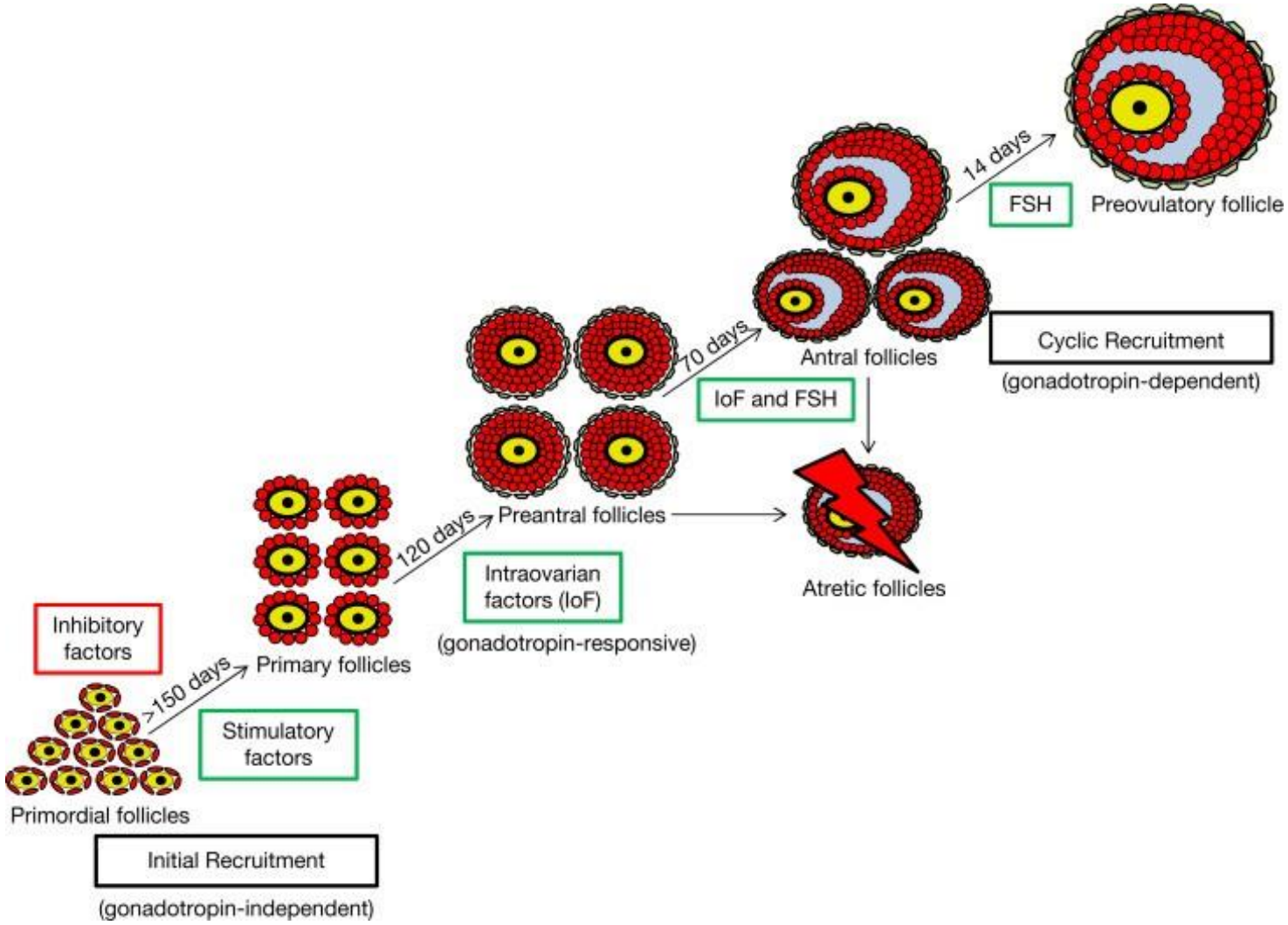


MATURE GRAAFIAN FOLLICLE



From Erickson et al

Follicular Development: Timeline



The follicle that is ovulated has been developing for >300 days

https://ars.els-cdn.com/content/image/3-s2.0-B9780128012383646287-f64628-04-9780128118993.jpg

Dominant Follicle

- Typically, each cycle only one graafian follicle becomes dominant and ovulates
  - Clinical correlate : some people have a genetic propensity to poly-ovulation. Those people may have fraternal twins run in their family. Identical twins are a random splitting event. Also it matters not at all if the sperm producing partner has twins in his family, it has only to do with poly-ovulation!
- Over a lifetime an estimated 400-500 follicles ovulate
  - ~40 years x 12 cycles/y
  - Decreased by pregnancy and ovulation preventing birth control
  - Pregnancy/contraception/breastfeeding cancel ovulation that month but don't "preserve" fertility
- The other few million undergo scheduled unavoidable atresia

Which Follicle Becomes Dominant?



- Poorly understood
- More vascular supply
  - By day 9 of cycle it has twice the vascular support of other antral follicles -> more cholesterol available, more gonadotrophin exposure
- More FSH in its follicular fluid and increased density of FSH receptors- not clear HOW this happens
- The dominant follicle is likely selected during the luteal phase of the <u>previous</u> menstrual cycle and grows rapidly until ovulation







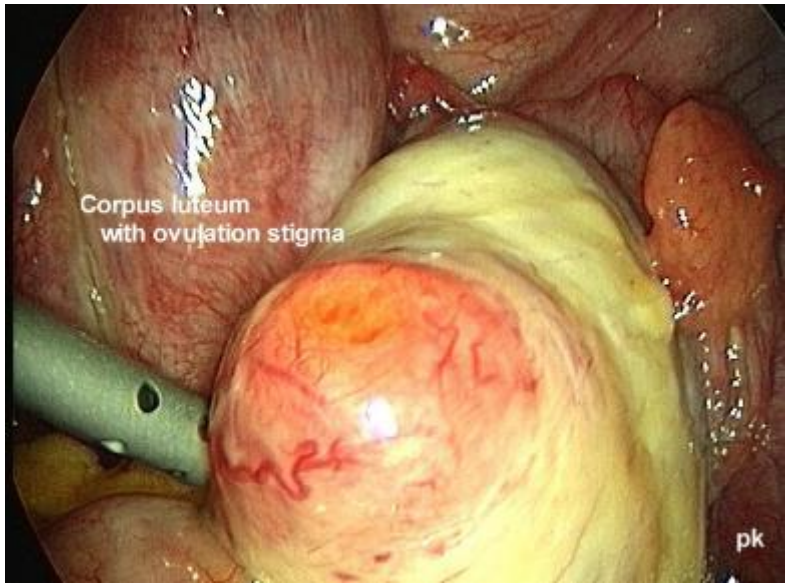


Ovulation

- Midcycle LH surge comes from rising levels of estradiol and midcycle switch in hypothalamus and pituitary to positive estradiol feedback
- The LH surge causes meiosis to restart and the egg becomes a "secondary oocyte"
- Granulosa cells transform into luteal cells (differentiation block is lifted)
- Proteolytic enzymes degrade the follicle wall
- Increased prostaglandin production
- All these events lead to ovulation

Corpus Luteum
"Yellow Body"

- Walls of the dominant follicle collapse
- Granulosa cells undergo hypertrophy and luteinization and begin to produce progesterone in addition to estradiol
- Forms a yellow cystic structure called the corpus luteum which is maintained by LH



Corpus Luteum
"Yellow Body" in Latin

How long does the corpus luteum last?

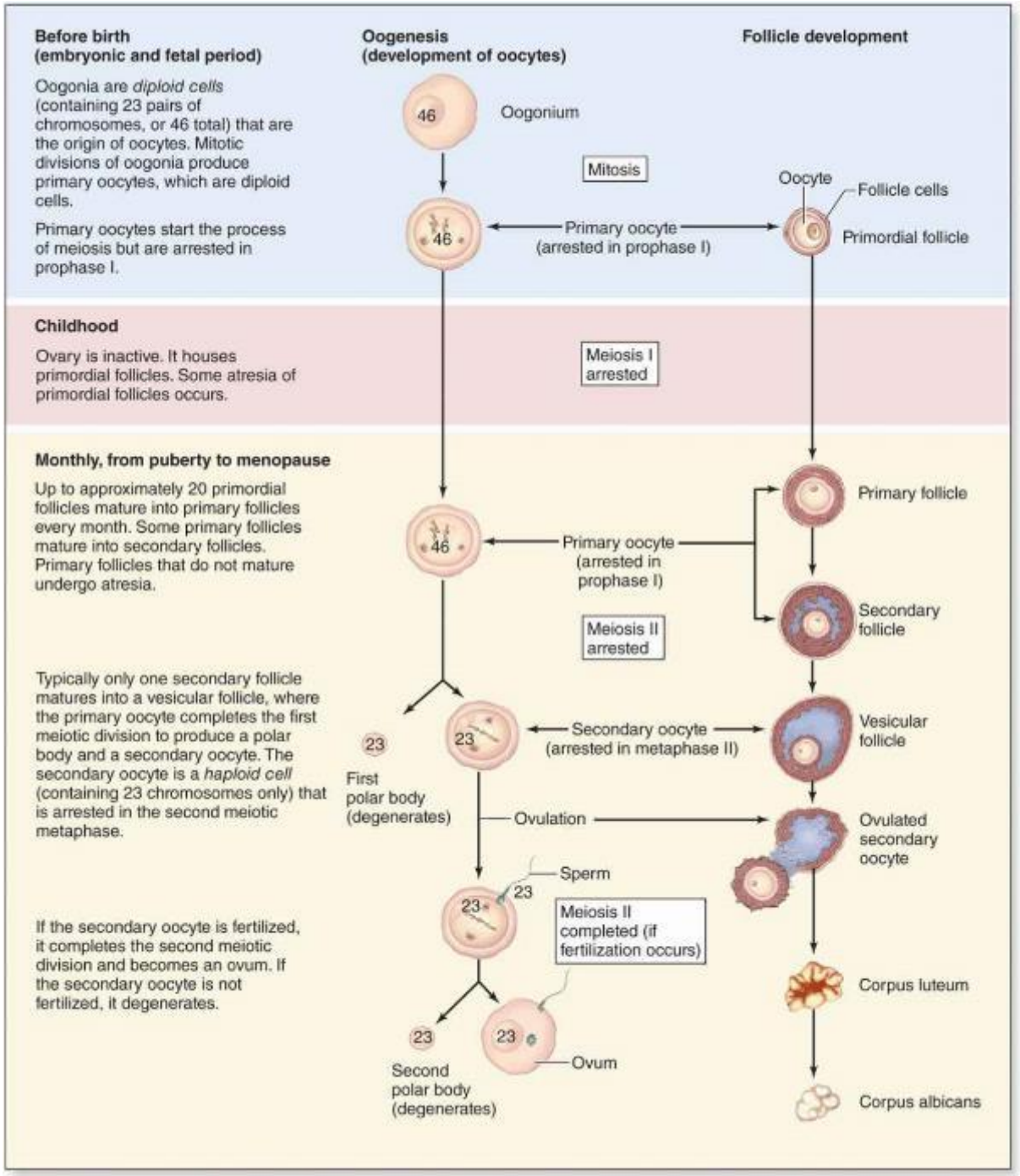
- If not pregnant--> 2 weeks
- Essential to maintaining pregnancy, if it occurs
- If not pregnant, corpus luteum regresses (becomes the corpus albicans) leading to fall of estradiol and progesterone levels
  - Uterus: Destabilization and shedding of the endometrium (menses)
  - Ovary: Increase in GnRH -> rise in FSH -> follicular growth

LO:

Summarize the process of meiosis in the oocyte

Meiosis

- Meiosis involves DNA replication (during the S phase) followed by two meiotic divisions
  - Take one diploid cell and make 4 unique daughter cells that are haploid (half the number of chromosomes as the parent cell)
  - In the case of the oocyte, it makes two polar bodies and one ovum
- All healthy oocytes in ovary are arrested in prophase of meiosis 1
- When exposed to surge of LH and FSH the oocyte restarts meiosis
- This allows for production of a mature gamete (and an extruded polar body)
- The prolonged arrest in Meiosis 1 is what predisposes to mis-segregation of chromosomes and therefore aneuploidy
  - Increasing age -> increasing time arrested -> increasing risk of aneuploidy due to chromosomal non-disjunction



- Primordial germ cells migrate to the fetal gonads and become oogonia
- Oogonia undergo mitotic expansion then enter meiosis 1 to become primary oocytes, then arrest in Prophase 1 until ovulation when meiosis resumes, and the first polar body is emitted
- It arrests again in metaphase 2, until fertilization and extrusion of the second polar body
- After fertilization, it is a zygote (egg containing two haploid nuclei- from egg and sperm- which then combine into a single diploid nucleus)- any genetic issues here will pass on to all cells
- Mitotic divisions grows cell number, and it becomes an embryo then eventually a fetus

Meiosis

Phase	What's happening?	Why should you care?
Prophase 1	Primary oocytes arrest here	Pairs recombine homologous recombination chromosomal crossover
Metaphase 1	Duplicate to make 2 identical chromatids	Line up at equator
Anaphase 1	Move in opposite directions	
Telophase 1	Unequal cell division	After LH surge, makes 1st polar body and secondary oocyte
Prophase 2	New spindle around chromosomes	
Metaphase 2	Line up again	Arrested here until ovulation occurs
Anaphase 2		If fertilization happens this last part happens
Telophase 2		

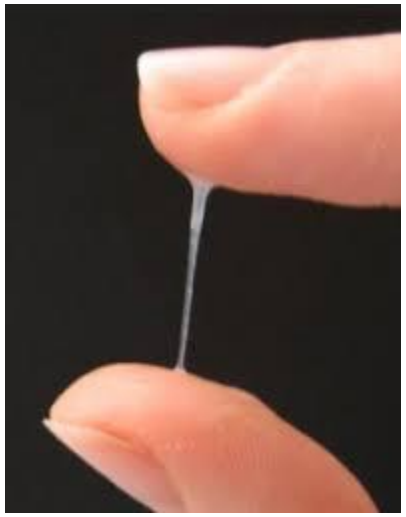
More detail in Dr. Spudich's reproductive embryology module!

LO:

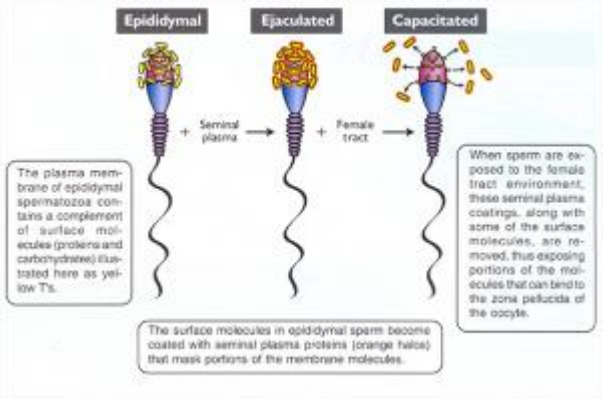
Describe the process of fertilization and implantation

Fertilization

- Sperm has to be present when ovum enters oviduct (uterine tube)
- Leading up to ovulation, the uterus and cervix facilitate sperm transport
  - Estrogen thins cervical mucus to allow sperm to pass
- Sperm undergoes capacitation in order to be able to penetrate the egg
  - Surface proteins removed
  - Acrosomal and plasma membranes united



Spinnbarkeit



ICSI

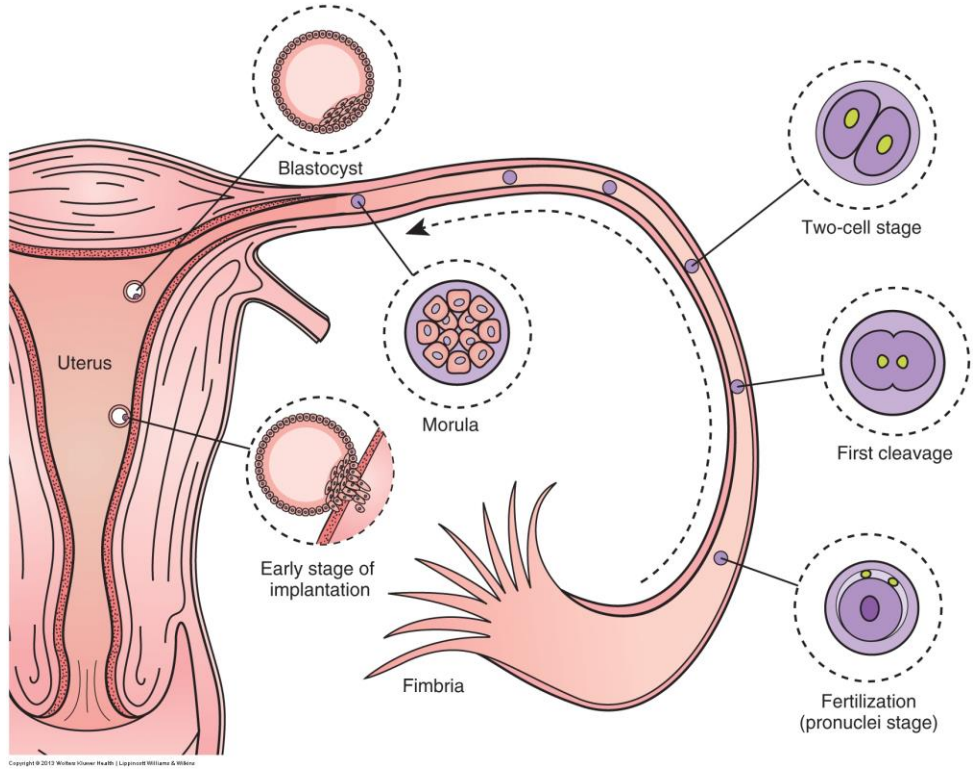

Fertility Associates
youtu.be/GTiKFCkPaUE
via /u/aloofloofah

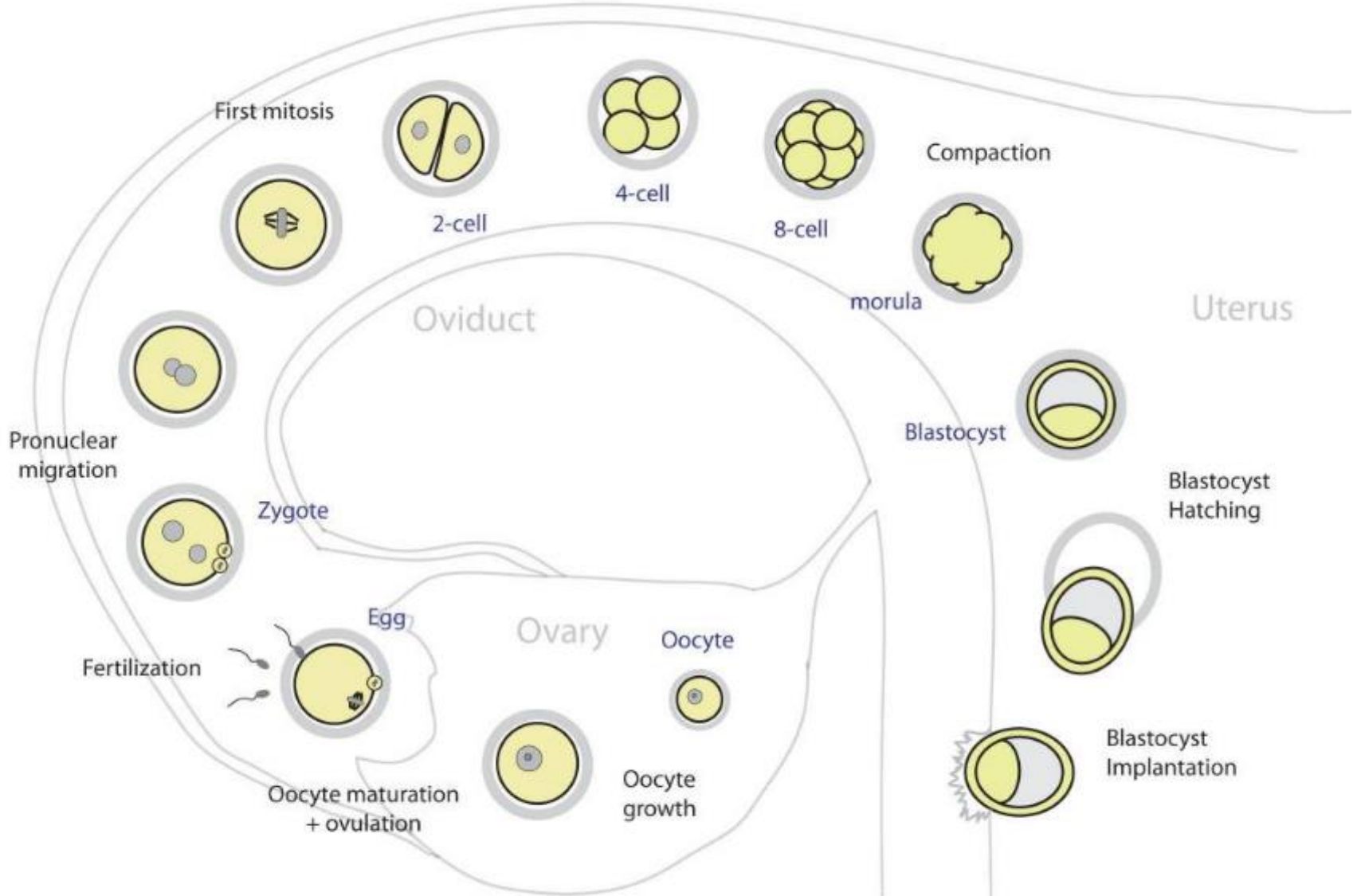
Embryo development video

- https://youtu.be/V6-v4eF9dyA

Once fertilized

- Travels through oviduct/fallopian tube for 2-3 days
- Day 3-4 blastocyst has formed, entered uterus
- Implantation occurs on Day 9-10 after ovulation





https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4021448/

LO:

Describe the physiology of sexual response and sexual dysfunction



Sexual Physiology





- Biology of human sexual activity is poorly understood
- Complex, can't use non-human animal data
- The human brain is more important than olfactory systems
- Limbic system (behavioral and emotional center) coordinates sex drive and mating
- Sacral spinal cord responsible for vaginal engorgement and orgasm









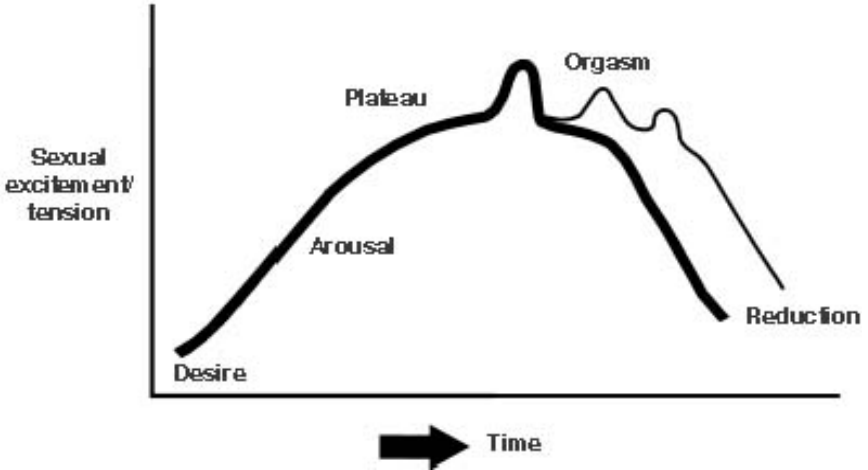


Sexual Response

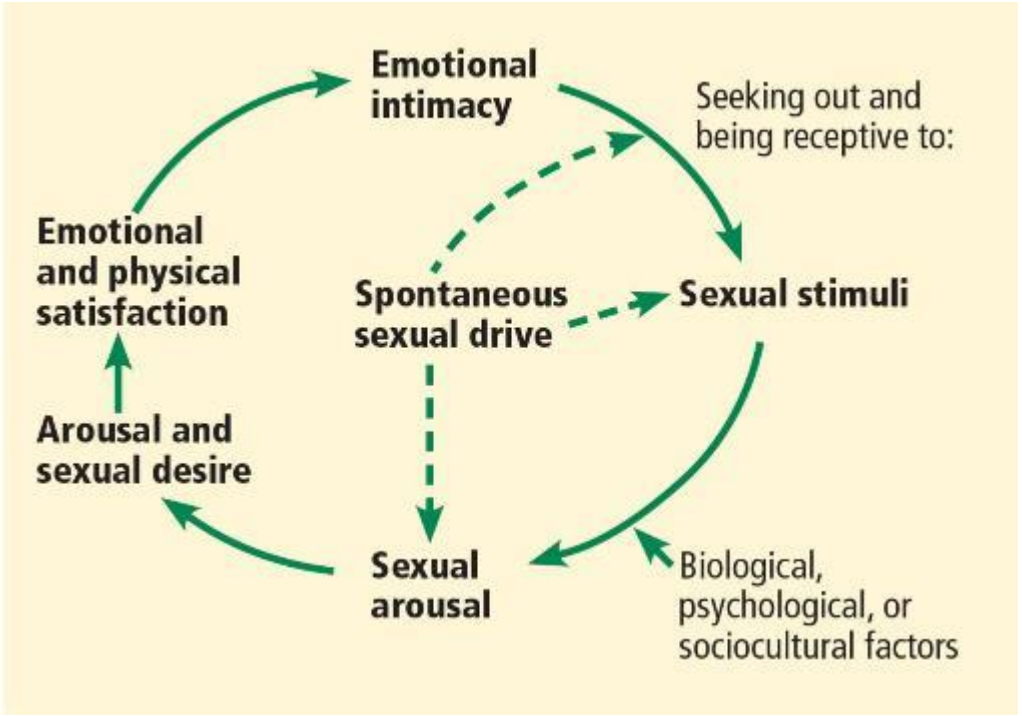
- Sexuality has a broad range of expressions of intimacy
- Strong cultural, religious, biologic/hormonal and psychological components
- Important quality of life issue (for most people, not all!)
- Must engage the body **and** the mind typically
- Up to 25-45% of people with ovaries perceive they have sexual dysfunction

Sexual Response Models

- Linear model
  - Desire- flushing, increased genital blood flow/swelling
  - Arousal- more swelling, tachycardia, muscle tension
  - Orgasm- contraction of the vaginal muscles
  - Resolution- return to normal functioning



- Intimacy Based Model



Sexual Response

- Many neurotransmitters and hormones have positive and negative effects on the sexual response cycle
  - Norepinephrine, dopamine, oxytocin and serotonin (5-HT), testosterone, estradiol
  - Prolactin, cortisol, GABA and serotonin via other receptors

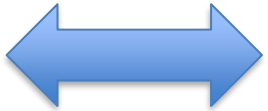
What can go wrong?

- Sexual dysfunction
  - Hypoactive sexual desire: most common; lack of interest
  - Arousal disorder: desire present, but arousal difficult to achieve or maintain
  - Orgasmic disorder: difficulty achieving orgasm
  - Vaginal pain disorder (dyspareunia, vaginismus)

- **Remember: it is only a problem if it is a problem for the person or their relationship**

What can go wrong?

- Causes of sexual dysfunction
  - Physical
    - Many medical conditions and the medications used to treat them can affect desire and function
  - Hormonal – low estrogen and/or androgen levels
    - Menopause/ovarian insufficiency
    - Postpartum
    - Lactation
  - Psychological
    - Untreated anxiety/depression, or the meds used to treat A/D
    - Conflict with your partner
    - History of sexual trauma/abuse



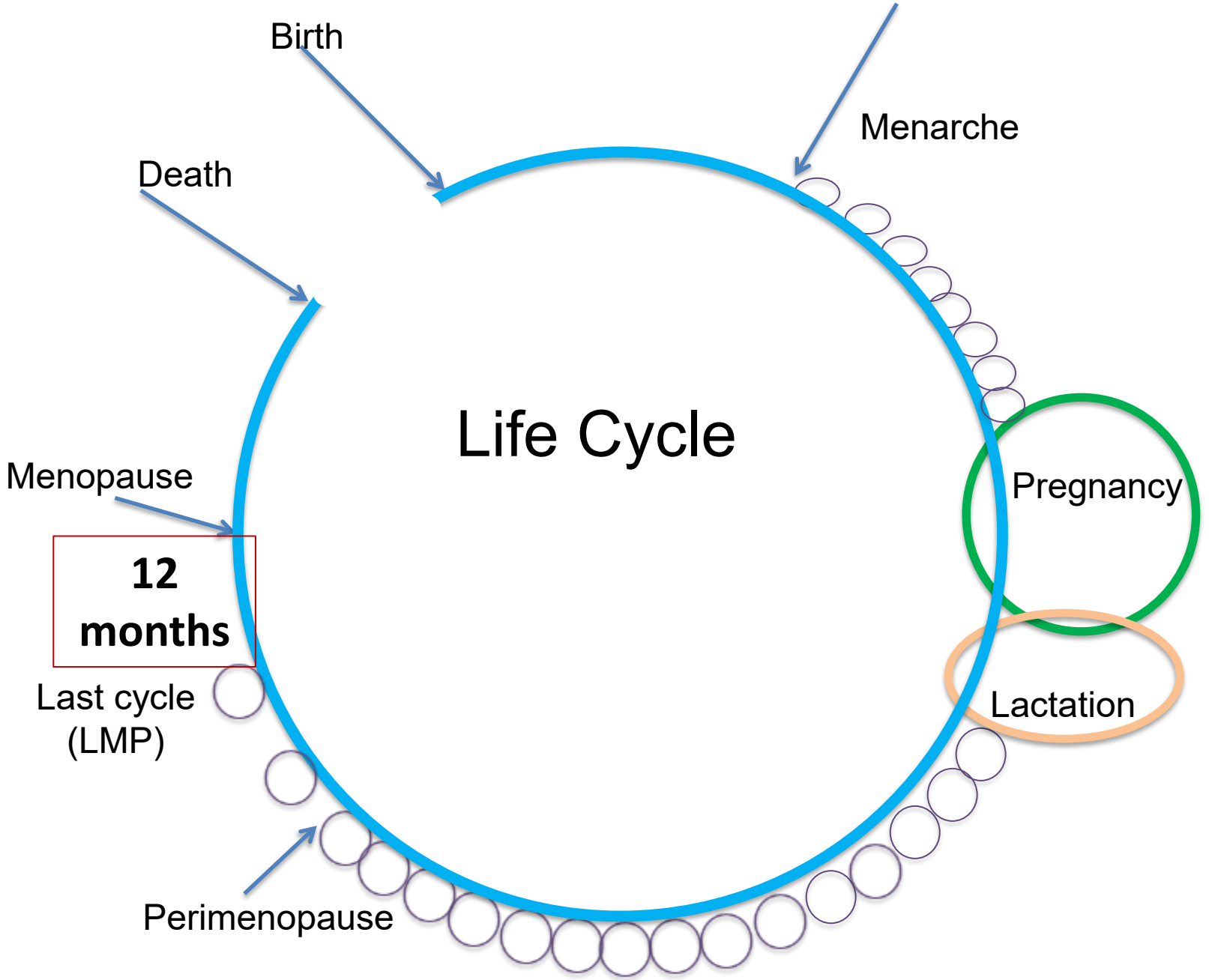
- Risk factors for sexual dysfunction
  - Depression or anxiety
  - Heart and blood vessel disease
  - Neurological conditions, such as spinal cord injury or multiple sclerosis
  - Gynecological conditions, such as vulvovaginal atrophy, infections or lichen sclerosus
  - Certain medications, such as antidepressants or high blood pressure medications
  - Emotional or psychological stress, especially with regard to your relationship with your partner
  - A history of sexual abuse

LO:

Define peri-menopause and menopause and describe the most commonly experienced hypoestrogenic symptoms

Peri-Menopause

- Before menopause
- Cycle becomes irregular
- Estrogen decreases
- Hypoestrogenic symptoms can begin years before the LMP
- Few healthy eggs left
- Progesterone decreases due to fewer ovulatory cycles
- Testosterone changes vary (mild decline to relatively stable)



Menopause

- Cessation of menses for 12 months
- No viable eggs left
- Very low estrogen and progesterone production; low testosterone
- Elevated FSH/LH levels

TABLE 37.3 Serum Gonadotropin and Steroid Levels in Premenopausal and Postmenopausal Women

		Menstrual Cycle			
Hormone	Units	Follicular	Preovulatory	Luteal	Postmenopausal
LH	mIU/mL	2.5–15	15–100	2.5–15	20–100
FSH	mIU/mL	2–10	10–30	2–6	20–140
Estradiol	pg/mL	70–200	200–500	75–300	10–60
Progesterone	ng/mL	≤0.5	≤1.5	4–20	≤0.5

LH, luteinizing hormone; FSH, follicle-stimulating hormone.

Symptoms of Hypoestrogenic State

- Vasomotor symptoms
  - Hot flashes/night sweats
- Insomnia
- Brain fog/concentration difficulty
- Changes in mood and patience level
- Hair loss
- Decreased libido
- Genitourinary Syndrome of Menopause (usually later)
  - Urinary symptoms
    - Urgency
    - Stress incontinence
  - Vaginal dryness
  - Dyspareunia (pain with sex)

Post-Menopausal Shifts in Progesterone and Testosterone

Progesterone

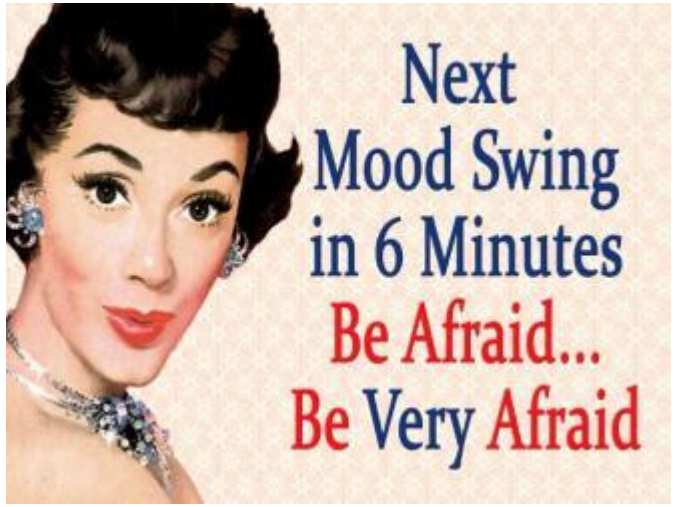
- Sharp decline due to cessation of ovulation
- Increased risk of endometrial hyperplasia (if unopposed estrogen)
- Sleep disturbances
- Mood variability

Testosterone

- Gradual decline from ovarian and adrenal sources
- Low libido
- Reduced lean muscle mass
- Reduced bone density
- Fatigue
- Cognitive changes

Treatment Options for Menopausal Symptoms

- Vasomotor symptoms: due to change in temperature sensation in the brain
  - HT (estrogen or estrogen + progesterone)
  - Fezolinetant- *New* - Neurokinin B blocker, works in hypothalamus
  - SSRIs and SNRIs - SSRIs and SNRIs increase serotonin levels, decreasing hyperthermia
  - Oxybutynin- anticholinergic used for OAB, blocks involuntary muscle activity
  - Gabapentin- antiepileptic, can also help with insomnia
  - Cognitive Behavioral Therapy
- Genitourinary syndrome: due to loss of estrogen effects on vagina and urethra- thinning of tissue and decreased self-lubrication
  - Topical estrogen therapy
- Mood changes
  - SSRI/SNRI first choice as they also help with hot flashes


Next
Mood Swing
in 6 Minutes
Be Afraid...
Be Very Afraid

The End


THE 7 MENOPAUSAL DWARFS
ITCHY
BITCHY
BLOATY
SLEEPY
SWEATY
FORGETFUL
PSYCHO
danc


The biggest lie I tell myself about menopause is "I don't need to write that down, I'll remember it."


LIFE WAS MOVING ALONG AT A NICE, STEADY PACE, THEN I HIT MIDDLE AGE AND MY LIFE HOT-FLASHED BEFORE MY EYES!


C'mon, you gotta have one more hot flash in you. That third step is still icy.
Blazek